

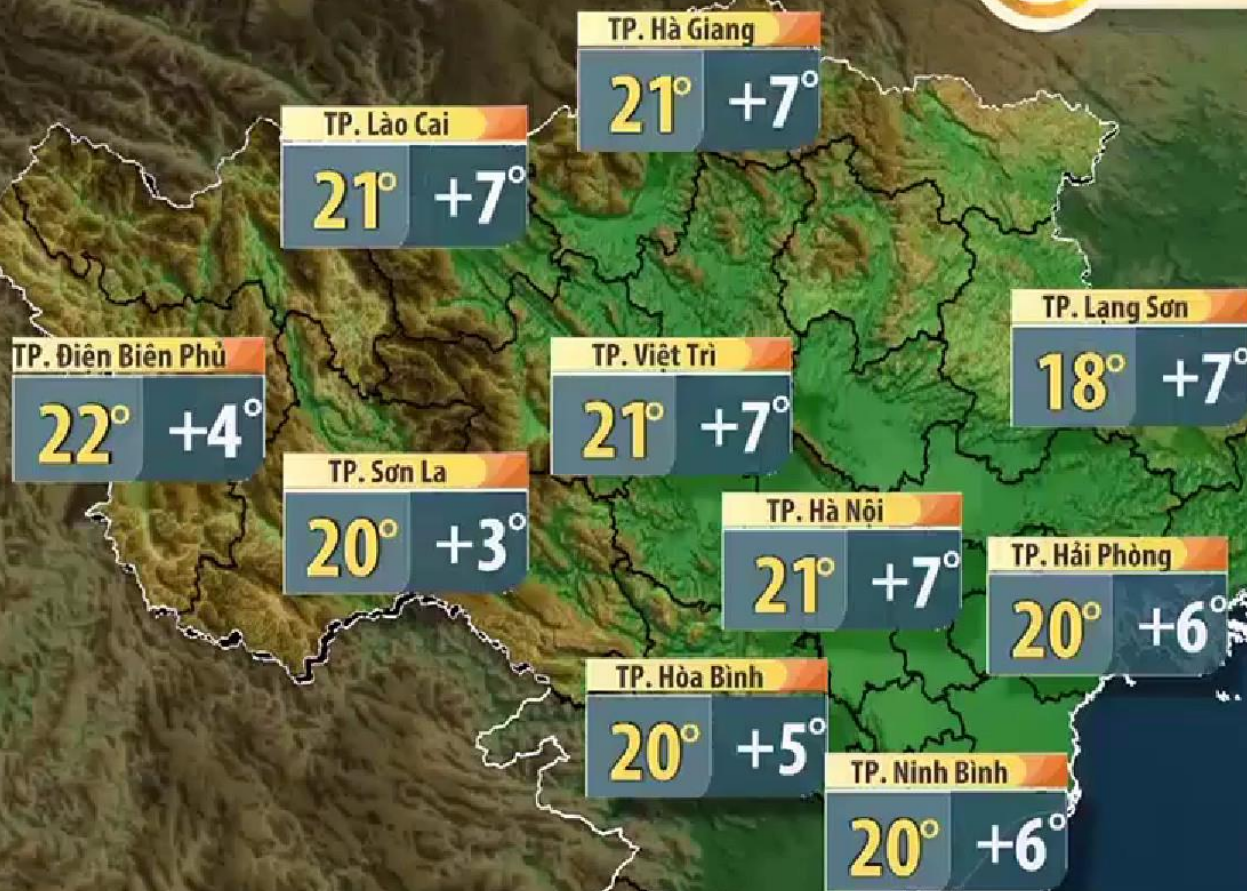
Introduction of Geographical Information System

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Chapter 3: Geospatial technologies

- 1. Map projection and coordinate system**
- 2. Land surveying**
- 3. Global navigation satellite systems**
- 4. Remote sensing**



Đ. Bạch Lồng Vĩ

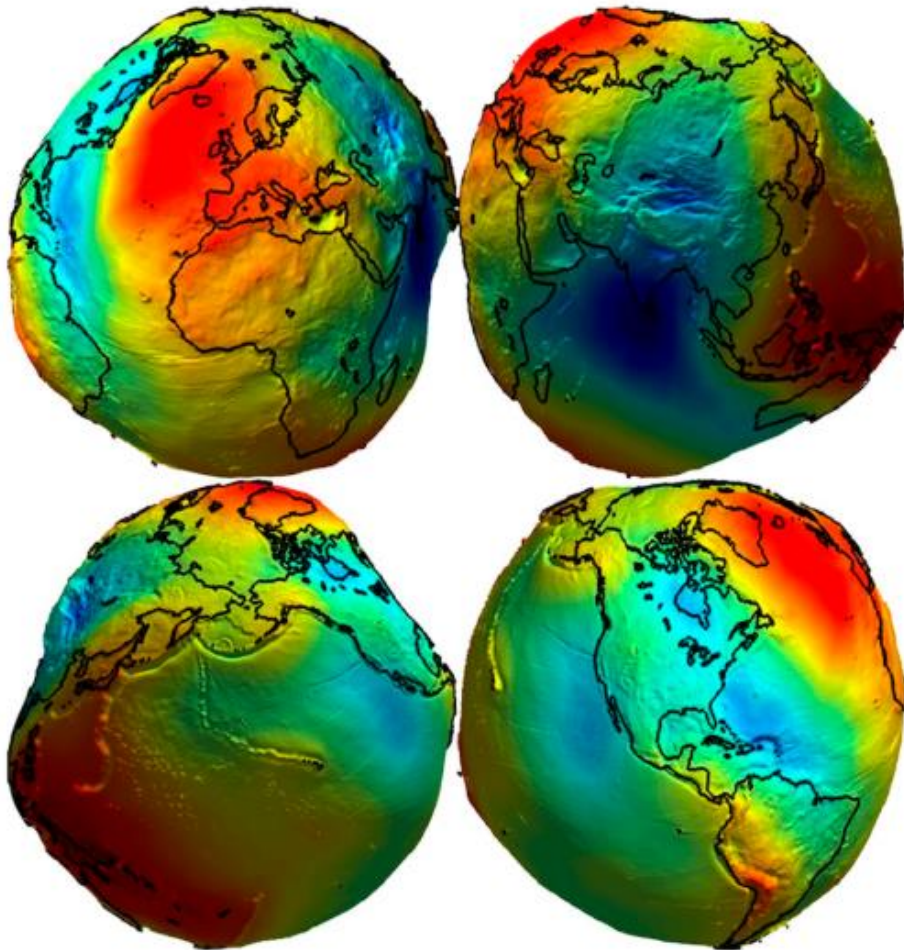
1. Map projection and coordinate system

- Geodesy – The shape of the Earth
- Geographic Coordinate Systems
- Datums
- Map Projections
- Projected Coordinate Systems

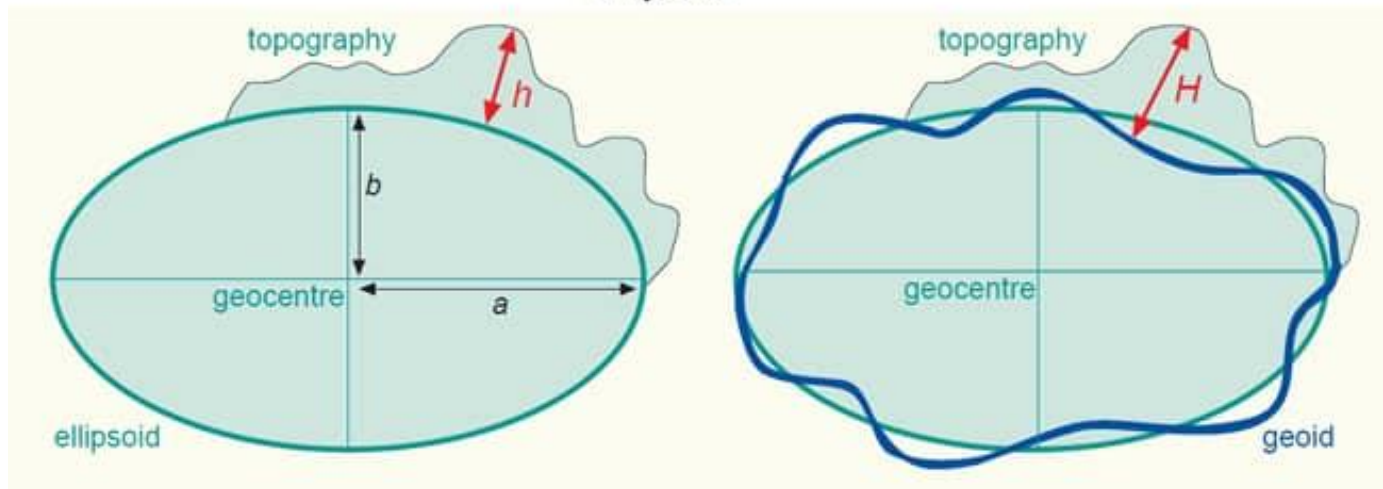
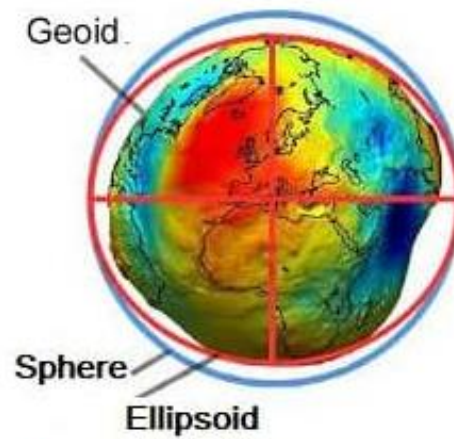
(Why do we need to know all definition?)

Earth models, Geoid?

What is the shape of Earth?



- Geoid: The planet's irregular topography and **mass distribution** (physical model)
- Geoid is calculated based on the hypothetical equipotential **gravitational surface**.
- The geoid approximates **mean sea level**. Height measured from geoid to the Earth surface is **Orthometric height**. Levelling gives orthometric height.
- **Can not apply** horizontal mathematical calculation
- used for **vertical measurements**
- True physical figure of the Earth to the idealized **geometrical figure** of a reference ellipsoid



Why Geoid Determination?

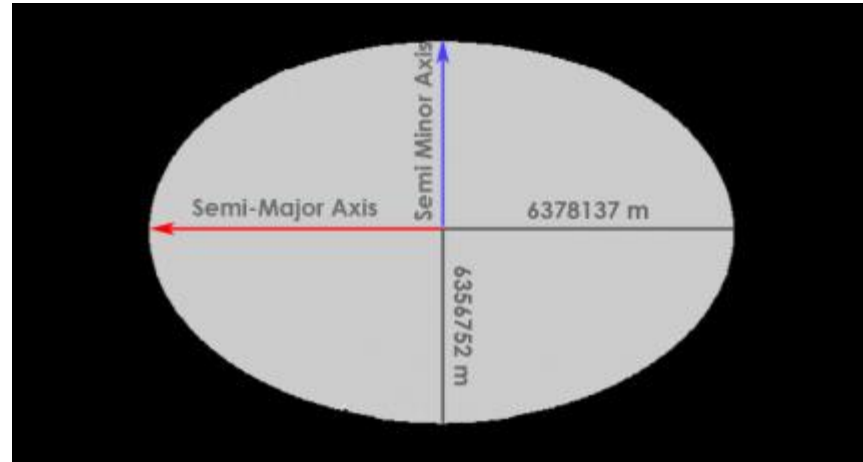
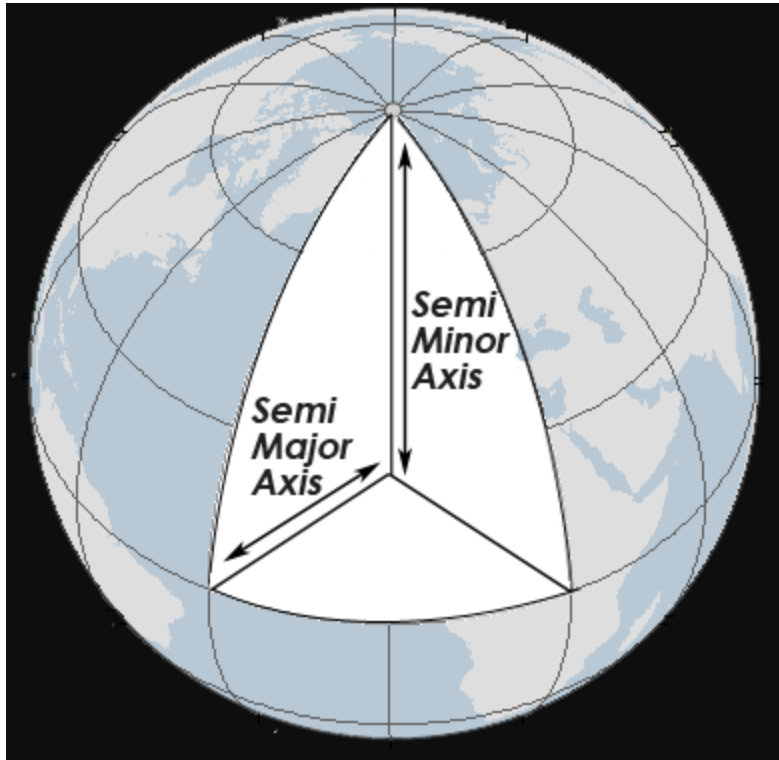
- Determine the **elevation** of any point with reference to the **reference geoid** at any time
- With the help of the Geoid we can complete the **construction** projects (that require elevation data) faster and in less cost.
- Height of any prominent **peaks** can be determined accurately (Ex. Solve the boundary conflicts).



Which one is higher?



Ellipsoid



WGS84 Ellipsoid:

Semi-major axis: 6,378,137.0 m

Semi-minor axis: 6,356,752.3 m

Inverse flattening: 294.978698214

Need a mathematical model

Ellipsoid: theoretical model of surface

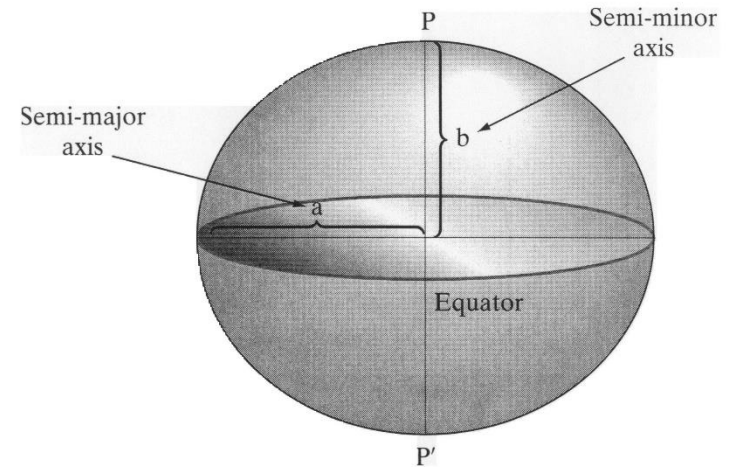
- not perfect sphere

- used for horizontal measurements

Ellipsoids: flattened spheres

- Degree of flattening given by $f = (a-b)/a$

(but often listed as $1/f$)

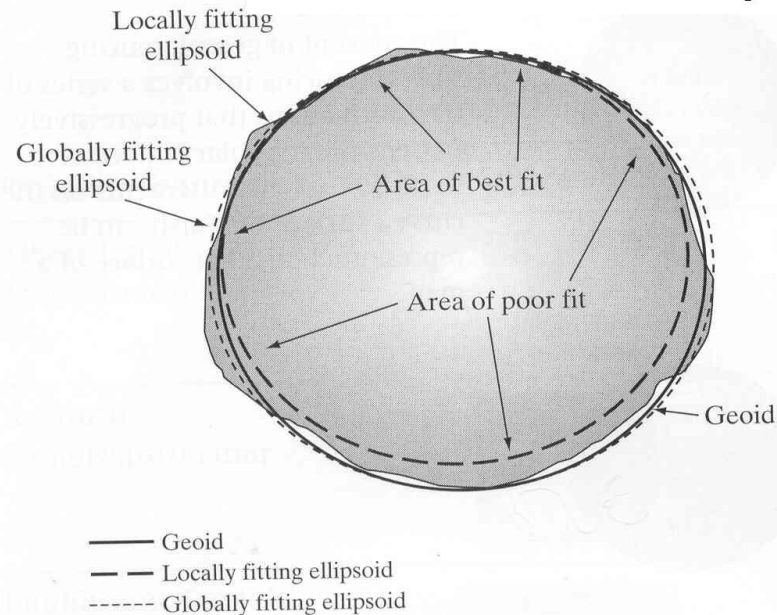


a = Major semi-axis of ellipsoid = Equatorial radius of the Earth

b = Minor semi-axis of ellipsoid = Polar radius of the Earth

PP' = Axis of revolution of Earth's ellipsoid

- Ellipsoid can be local or global



Examples of ellipsoids

Local Ellipsoids

Inverse flattening (1/f)

Airy 1830
Australian National
Clarke 1866
Clarke 1880
Everest 1956

299.3249646
298.25
294.9786982
293.465
300.8017

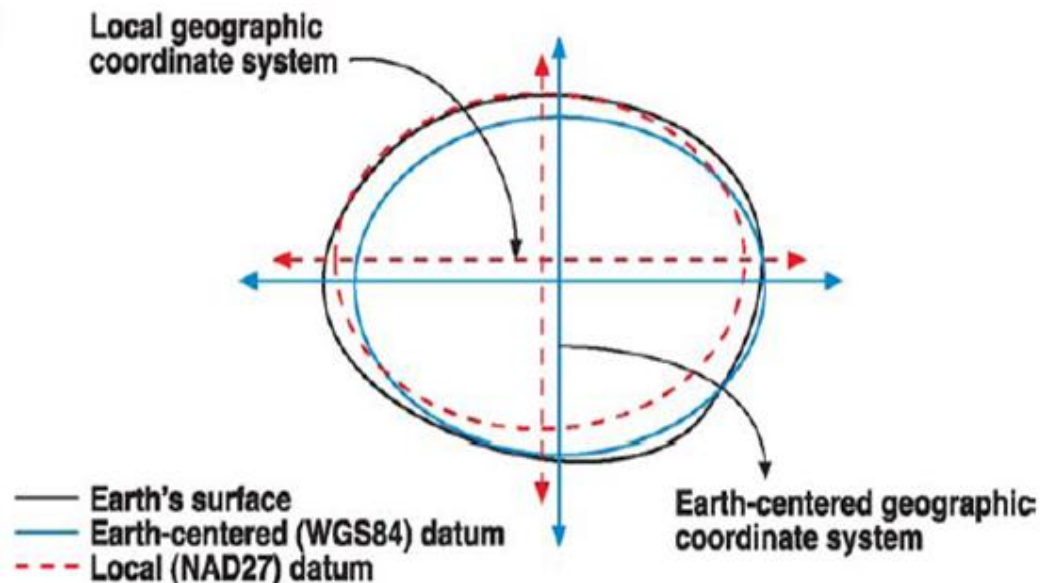
Global Ellipsoids

International 1924
GRS 80 (Geodetic
Ref. Sys.)

297
298.257222101

WGS 84 (World
Geodetic Sys.)

298.257223563



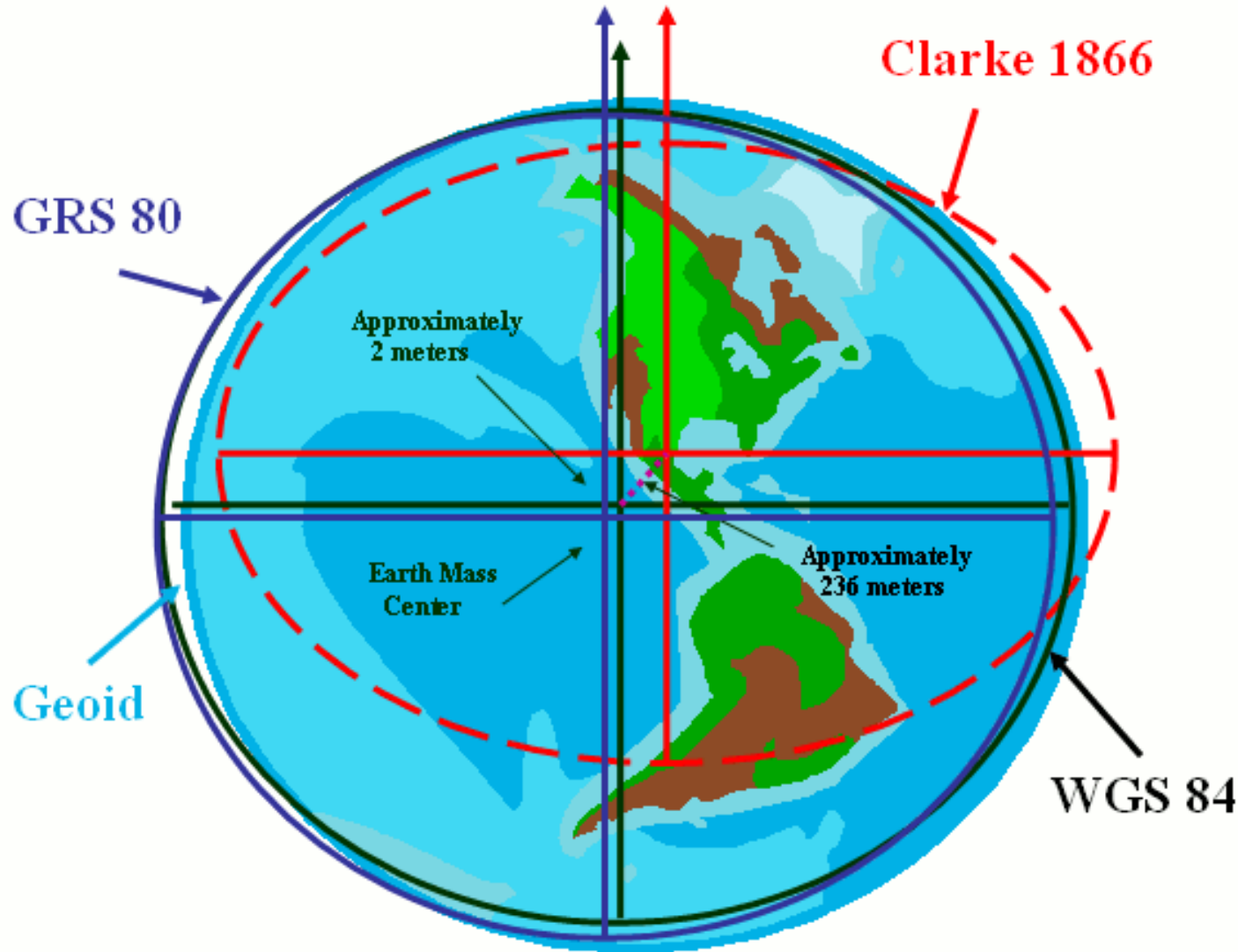
Historical determinations of the Earth's radius and flattening

*Flattening denoted by f .

author	year	method	equatorial radius (in metres)	$1/f^*$
P. Bouguer and P.-L. M. de Maupertuis	1735–43	arc	6,397,300	216.80
G.B. Airy	1830	arc	6,376,542	299.30
A.R. Clarke	1866	arc	6,378,206	295.00
F.R. Helmert	1884	gravimetric		299.25
J.F. Hayford	1906	arc	6,378,283	297.80
W.A. Heiskanen	1928	gravimetric		297.00
H. Jeffreys	1948	arc	6,378,099	297.10

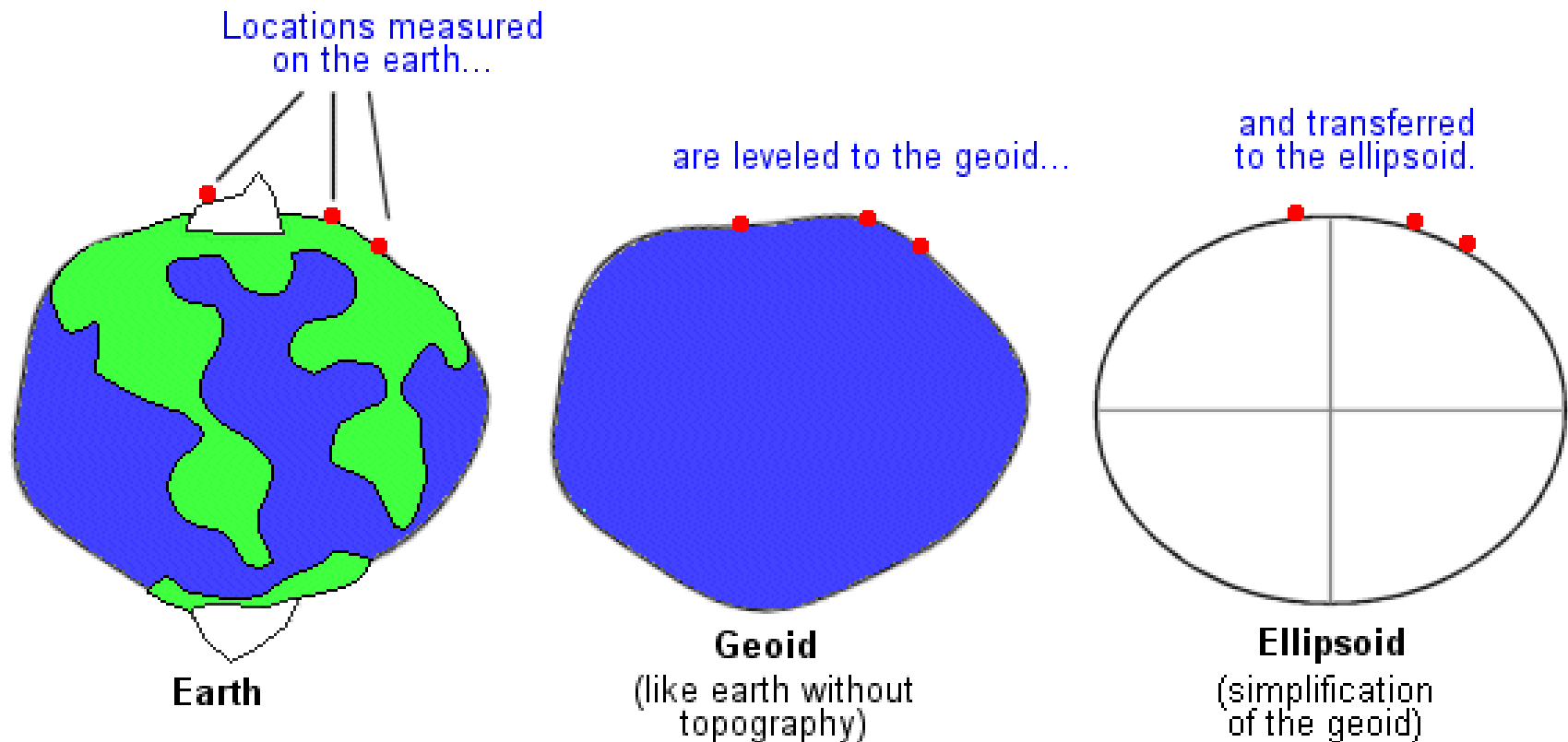
Ellipsoid Models

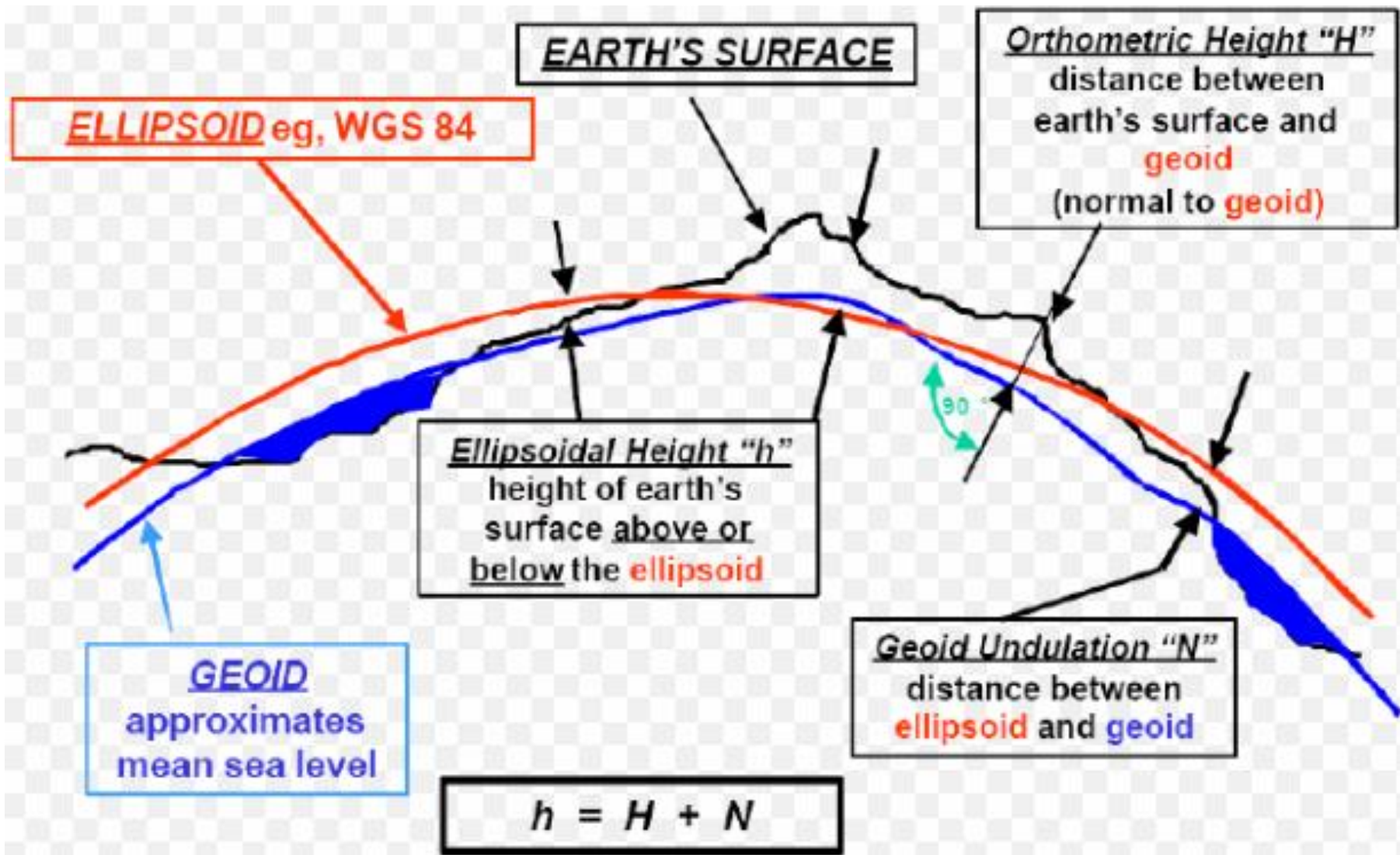
- Provide a reference for the **datum** and contribute to the datum's definition



- Requires survey measurements to define the shape of the earth
- Clarke 1866 Ellipsoid became the standard reference model for NAD27
- Geodetic Reference System of 1980 is the figure for NAD83

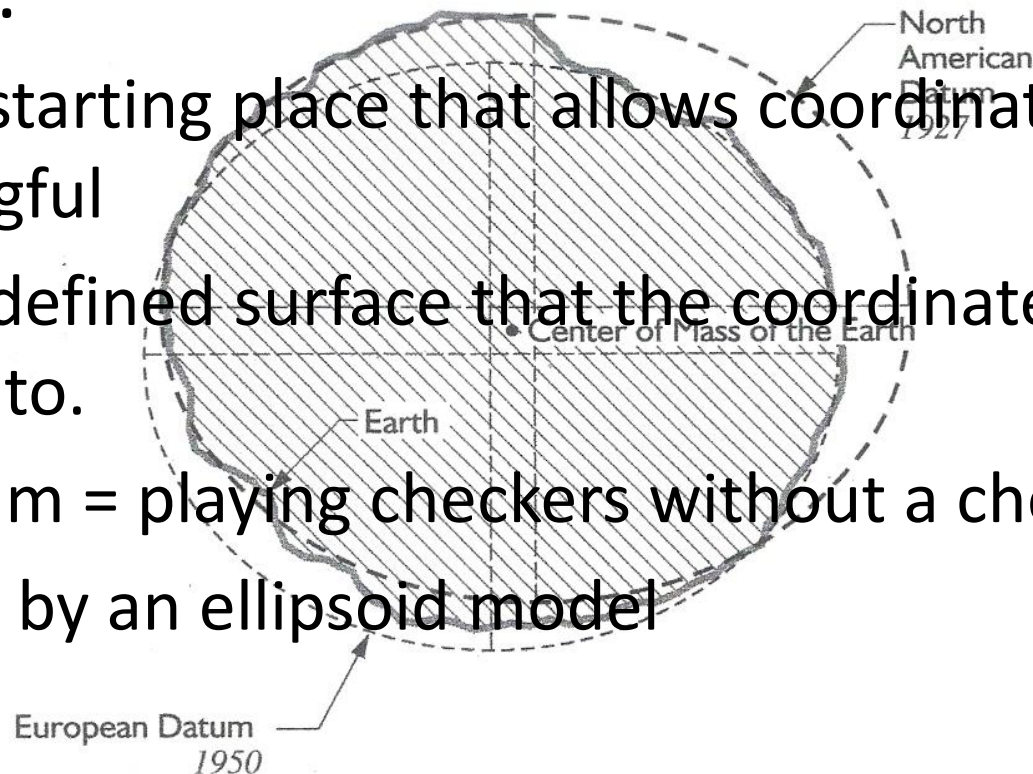
Models of the Earth





Datum

- Defines the size, shape, and orientation of Earth
- Provides..
 - Origin: starting place that allows coordinates to be meaningful
 - Clearly defined surface that the coordinate system is applied to.
 - No datum = playing checkers without a checkerboard
 - Defined by an ellipsoid model



Datum

- Datum: a set of reference measurements for a particular region, based on specified ellipsoid + geodetic control points
- > 100 world wide: NAD1983, WGS84, VN2000

Adindan	Adindan- Ethiopia, Mali, Senegal, Sudan
Algooye	Algooye- Somalia
AIN EL ABD '70	AIN EL ANBD 1970- Bahrain Island, Saudi Arabia
Anna 1 Ast '65	Anna 1 Astro '65- Cocos I.
ARC 1950	ARC 1950- Botswana, Lesotho, Malawi, Swaziland, Zaire, Zambia RC 1960 Kenya, Tanzania
Ascnsn Isld '58	Ascension Island '58- Ascension Island
Astro B4 Sorol	Sorol Atoll- Tern Island
Astro Bcn "E"	Astro Beacon "E"- Iwo Jima
Astro Dos 71/4	Astro Dos 71/4- St. Helena
Astr Stn '52	Astronomic Stn '52- Marcus Island
Aus Geod '66	Australian Geod '66- Australia, Tasmania Island
Aus Geod '84	Australian Geod '84- Australia, Tasmania Island
Austria	Austria
Bellevue (IGN)	Efate and Erromango Islands
Bermuda 1957	Bermuda 1957- Bermuda Islands
Bogata Observ	Bogata Observatry- Colombia
Campo Inchspe	Campo Inchauspe- Argentina
Canton Ast '66	Canton Astro 1966- Phoenix Islands
Cape	Cape- South Africa
Cape Canavrl	Cape Canaveral- Florida, Bahama Islands
Carthage	Carthage- Tunisia
CH-1903	CH 1903- Switzerland
Chatham 1971	Chatham 1971- Chatham Island (New Zealand)
Chua Astro	
Chua Astro- Paraguay	
Corrego Alegr	Corrego Alegre- Brazil
Djakarta	Djakarta (Batavia)- Sumatra Island (Indonesia)
Dos 1968	Dos 1968- Gizo Island (New Georgia Islands)
Dutch	Dutch
Easter Isld 67	Easter Island 1967
European 1950	European 1950- Austria, Belgium, Denmark, Finland, France, Germany, Gibraltar, Greece, Italy, Luxembourg, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland
European 1979	European 1979- Austria, Finland, Netherlands, Norway, Spain, Sweden, Switzerland
Finland Hayfrd	Finland Hayford- Finland

Appendix D

Map Datums

Gandajika Base	Gandajika Base- Republic of Maldives
GDA	Geocentric Datum of Australia
Geod Datm '49	Geodetic Datum '49-New Zealand
Guam 1963	Guam 1963- Guam Island
Gux 1 Astro	Guadalcanal Island
Hjorsey 1955	Hjorsey 1955- Iceland
Hong Kong '63	Hong Kong
Hu-Tzu-Shan	Taiwan
Indian Bngldsh	Indian- Bangladesh, India, Nepal
Indian Thailand	Indian- Thailand, Vietnam
Indonesia 74	Indonesia 1974-Indonesia
Ireland 1965	Ireland 1965- Ireland
ISTS 073 Astro	ISTS 073 ASTRO '69- Diego Garcia
Johnston Island	Johnston Island NAD27 Central
Kandawala	Kandawala- Sri Lanka
Kerguelen Isld	Kerguelen Island
Kertau 1948	West Malaysia, Singapore
L. C. 5 Astro	Cayman Brac Island
Liberia 1964	Liberia 1964- Liberia
Luzon Mindanao	Luzon- Mindanao Island
Luzon Philippine	Luzon- Philippines (excluding Mindanao Isl.)
Mahe 1971	Mahe 1971- Mahe Island
Marco Astro	Marco Astro- Salvage Isl.
Massawa	Massawa- Eritrea (Ethiopia)
Merchich	Merchich- Morocco

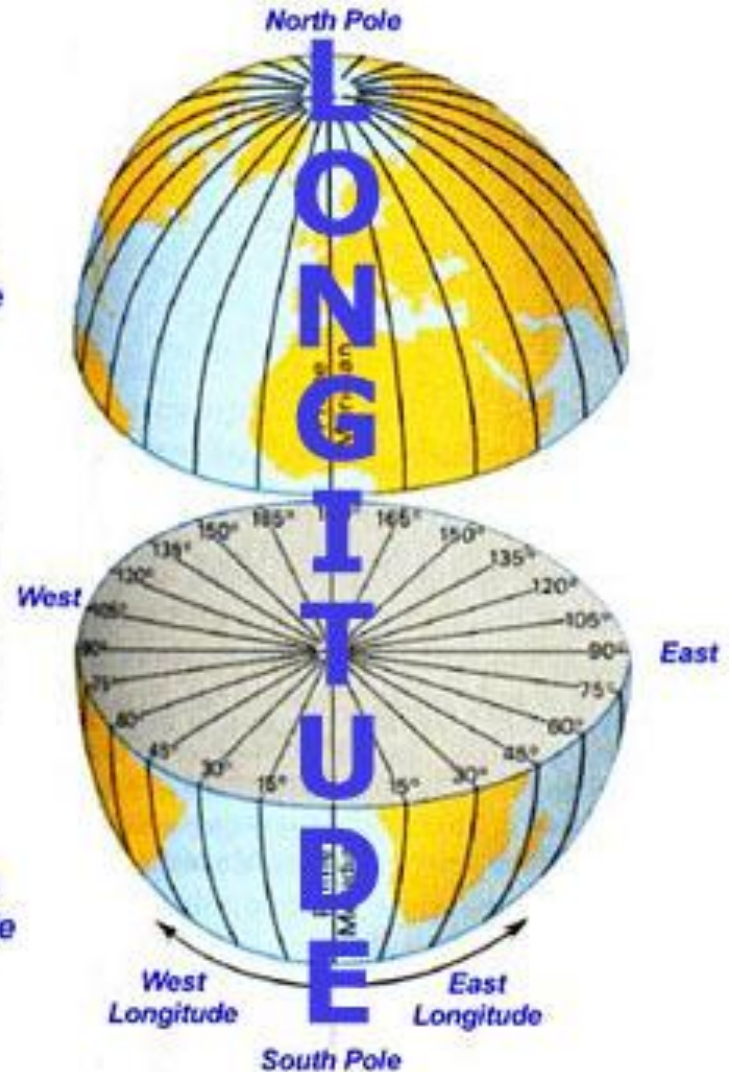
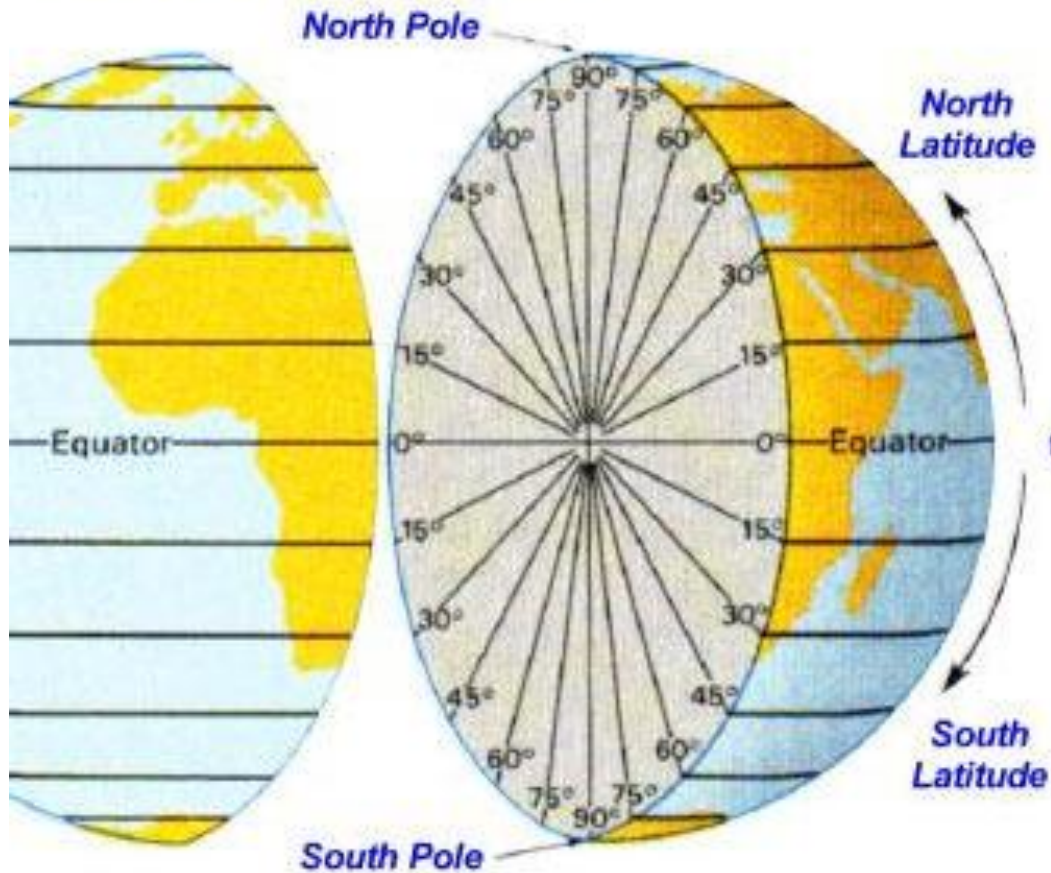
Some of the datums stored
in Garmin 76 GPS receiver

- **Coordinate System**- A reference framework consisting of:
A set of points, lines and/or surfaces, and a set of rules, used to define the **positions** of points **in space**, in either two or three dimensions.
- **Geographic Coordinate System** – A reference system that uses:
Latitude and longitude to define the locations of points on the surface of a **sphere or spheroid**.
A geographic coordinate system definition includes a **datum, prime meridian**, and angular unit of measurement.

Geographic Coordinate System

Latitude and longitude refer to degree, minutes and seconds of arc from reference lines that run East-West (latitude; equator) or North-South (longitude; prime meridian)

LATITUDE

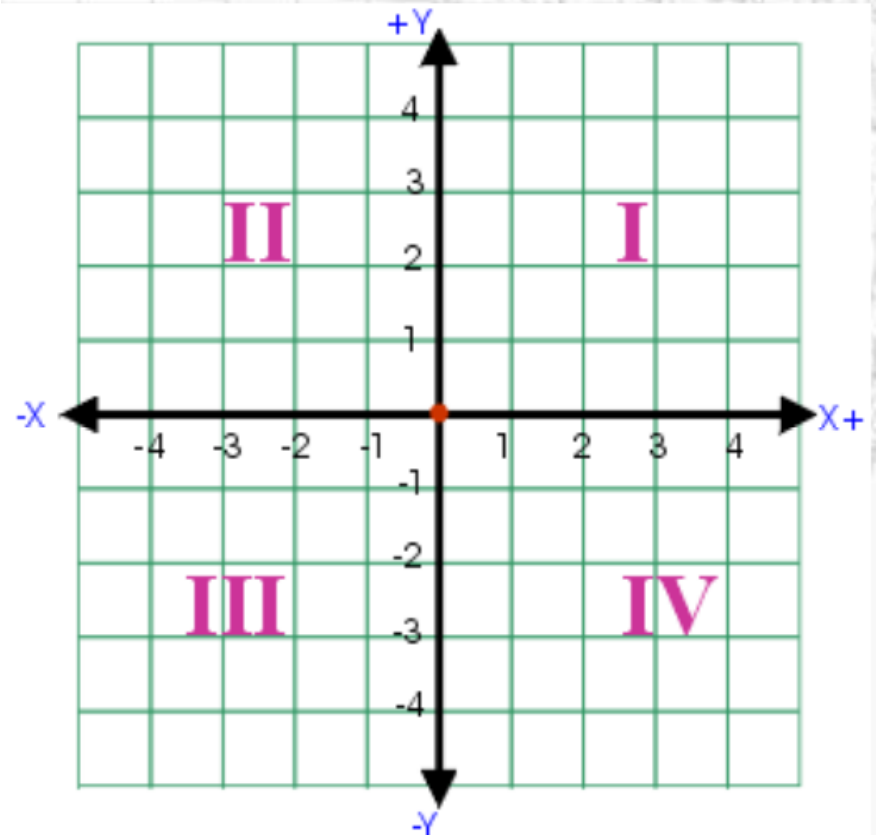


- **Projected Coordinate System** – A reference system used to: locate **x, y, and z** positions of point, line, and area features in **two or three dimensions**.
- A projected coordinate system is defined by a **geographic coordinate system**, a **map projection**, any **parameters** needed by the map projection, and a **linear unit** of measure.

- **Cartesian Coordinate System** – A two-dimensional, planar coordinate system in which **horizontal distance** is measured along an x-axis and vertical distance is measured along a y-axis.

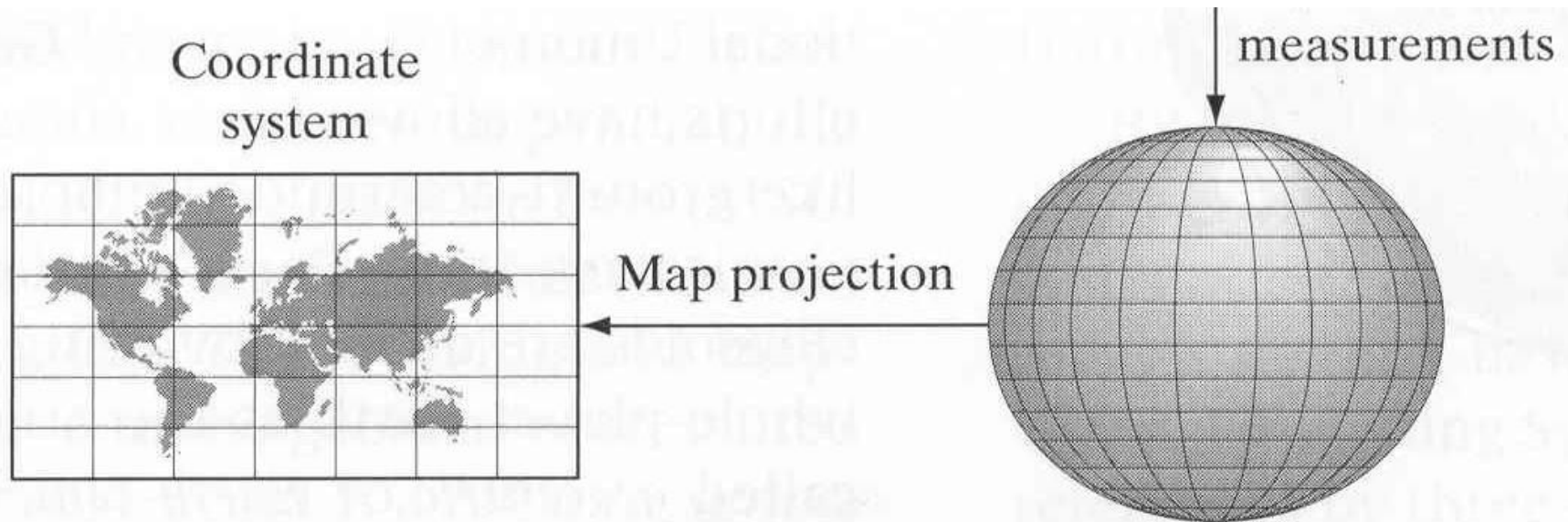


René Descartes, 1596-1650



Map projections

A reminder: the Earth is not flat!



Producing a perfect map projection is like peeling
an orange and **flattening the peel** without **distorting**
a map drawn on its surface.

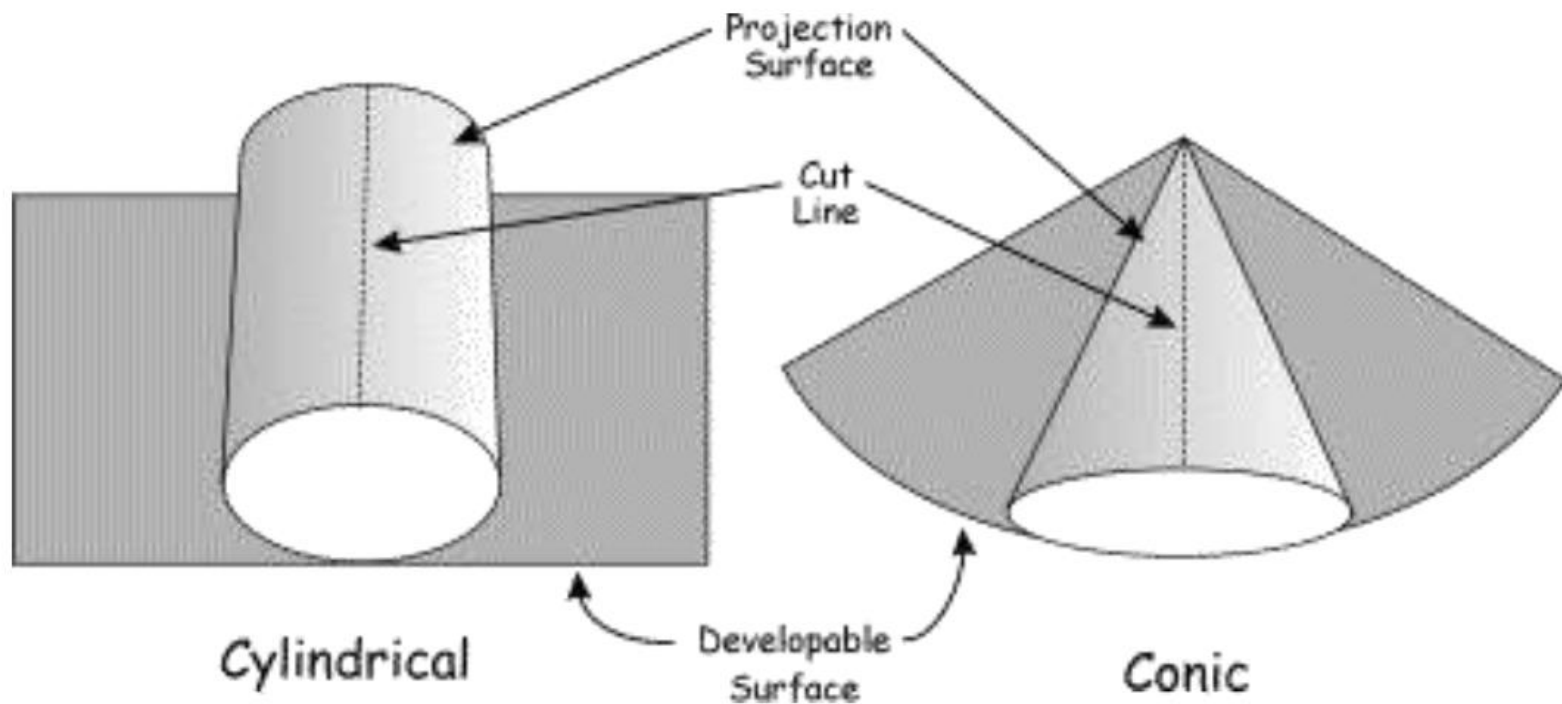
Projection

(Material by Rowan University)

- **Projecting** is the science of converting the **spherical earth surface** to a **flat plane**
- No system can do this perfectly. Some **distortion** will always exist.
- Properties that are distorted are **angles, areas, directions, shapes and distances.**
- Each projection **distorts one or** more of these while **maintaining** others.
- Selecting a projection is based on selecting **which needs** to be preserved.

Developable Surfaces

- A surface that can be **made flat** by cutting it along **certain lines** and unfolding or unrolling it.
Cones, Cylinders, & Planes are all developable surfaces.



Developable Surfaces

- Projection families are based on the **developable surface** that is used to create them
 - Cones = conical projections
 - Cylinders = cylindrical projections
 - Planes = azimuthal or planar projections

Properties of a map projection

- Area
- Shape
- Distance
- Direction

Projections that conserve
area are called
equivalent

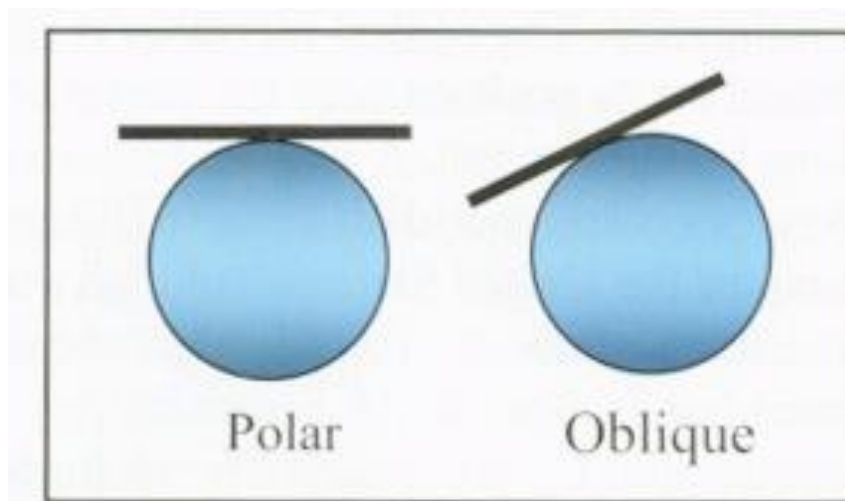
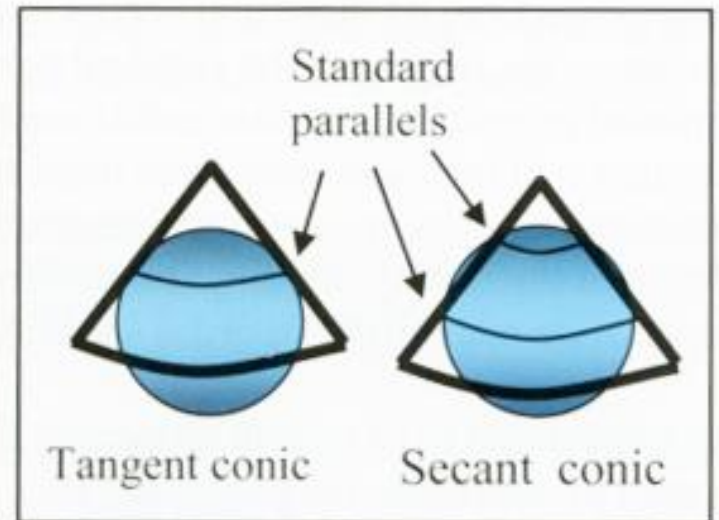
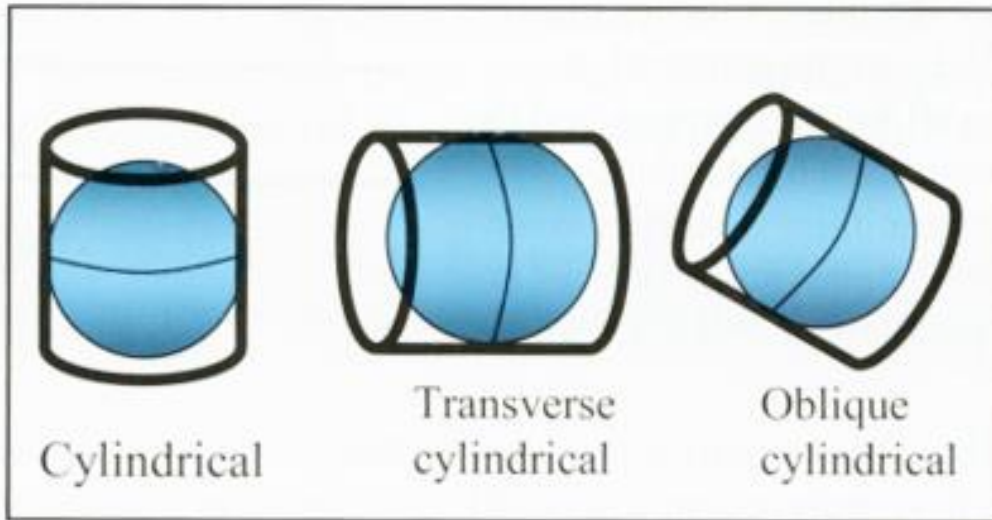
Projections that conserve
shape are called
conformal

Two rules:

Rule #1: No projection can preserve **all four properties**.
Improving one often makes another worse.

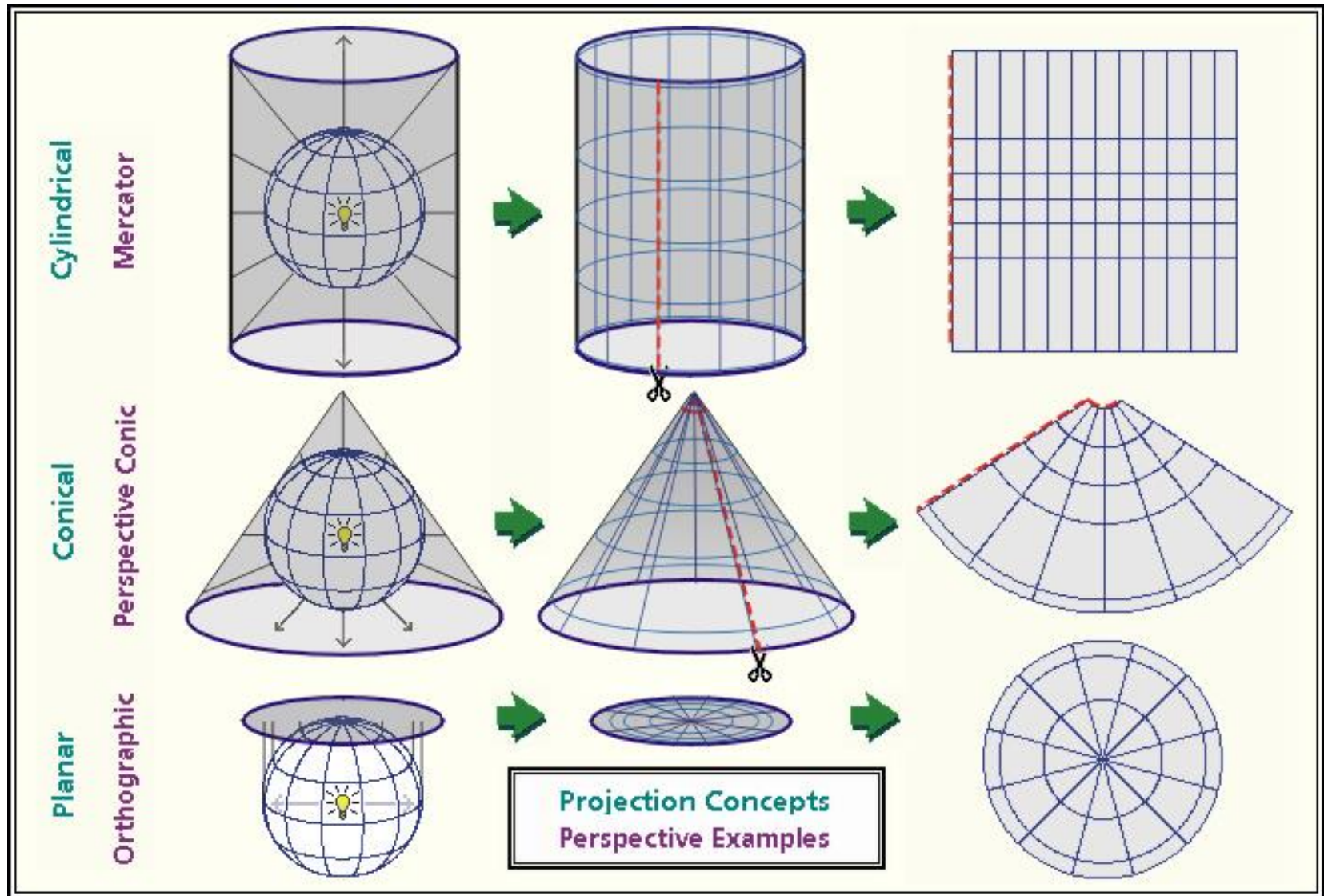
Rule #2: Data sets used in a GIS must be in **the same projection**.
GIS software contains routines for changing projections.

Projection types used in making maps



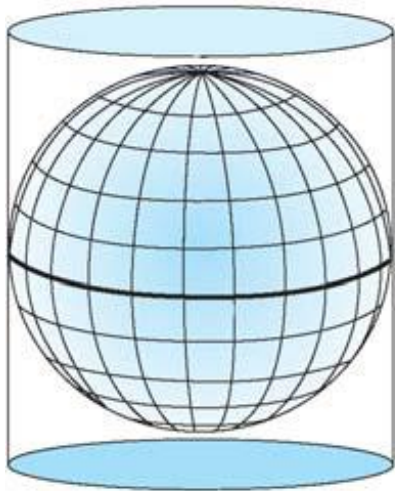
Map projections

Think of this as having a light in the middle of the earth, shining through the earth's surface, onto the projection surface



Cylindrical Projections

Different lines are tangential, but directions are true



Normal

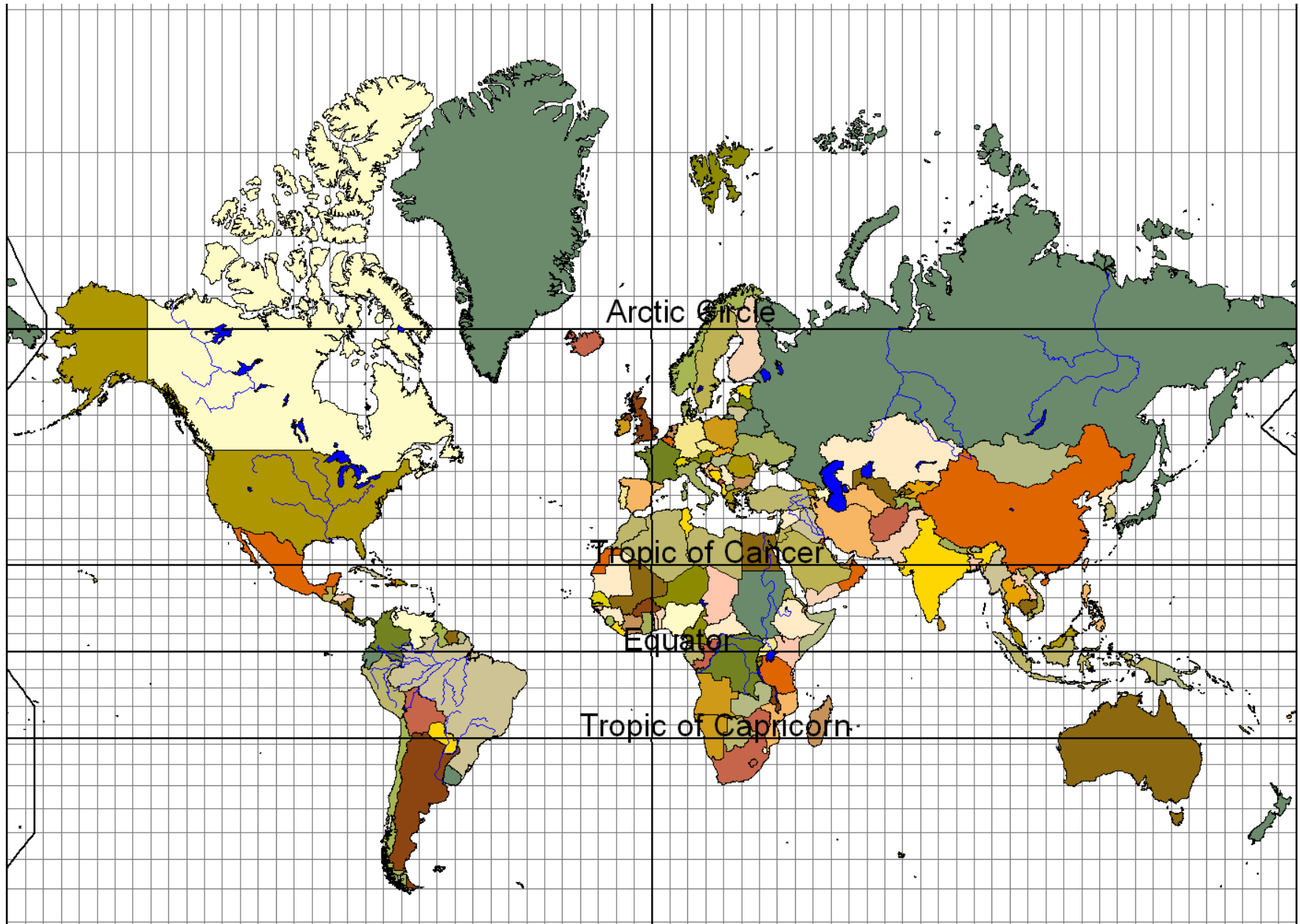


Transverse



Oblique

Cylindrical Projections

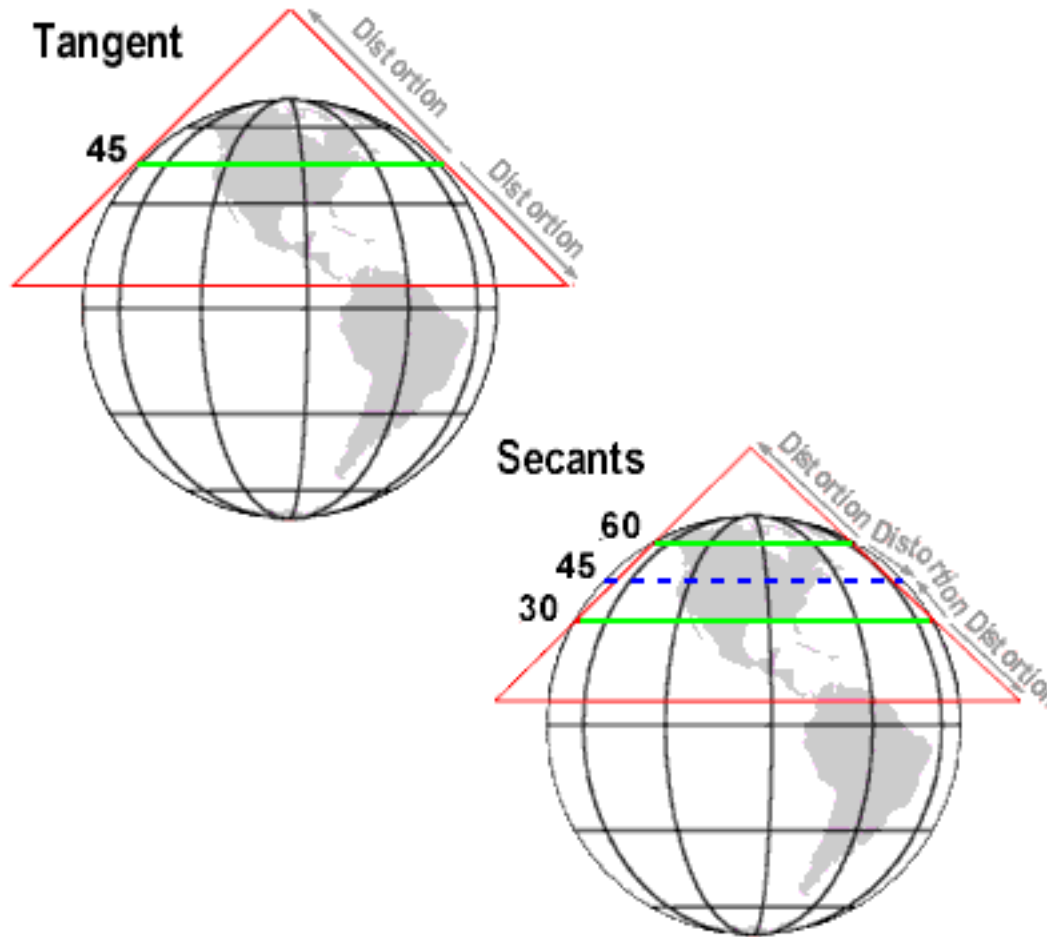


Cylindrical Projections

(conserve shape)

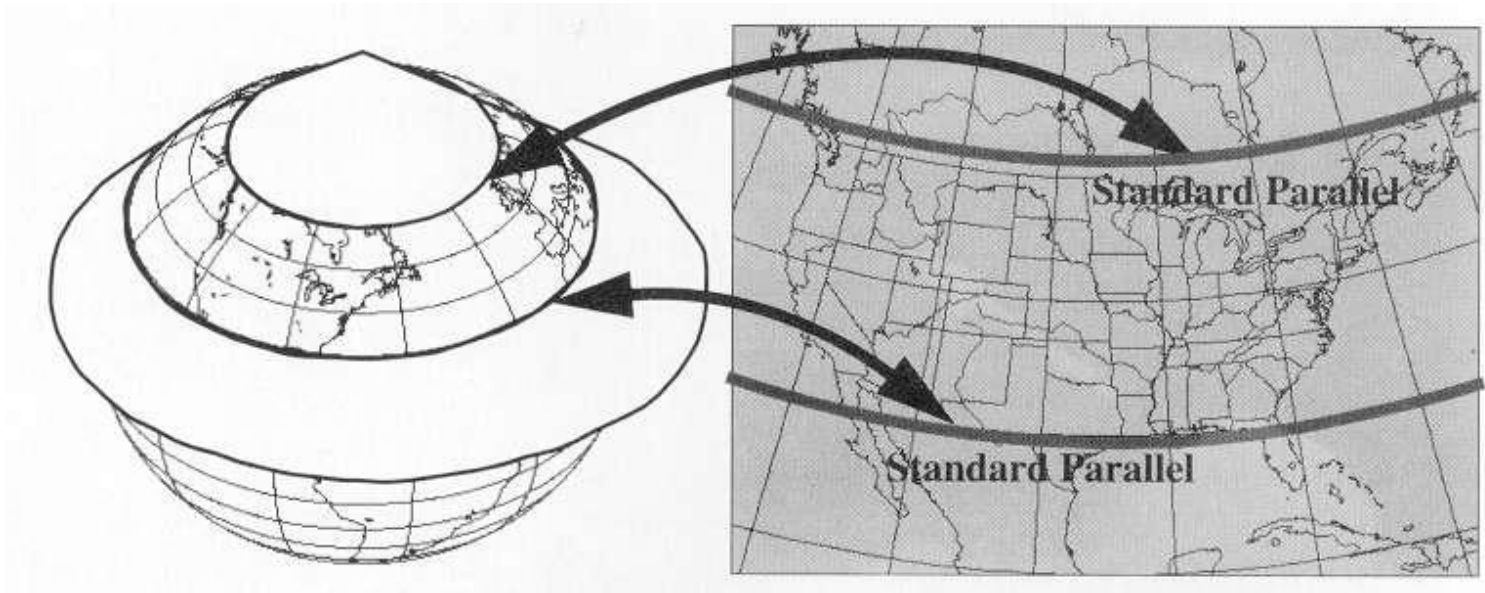
- Retain correct **angular relations** in transfer from globe to map.
- Angles correct for small areas.
- Scale same in any direction around a point, **but scale changes** from point to point.
- Parallels and meridians cross at right angles.
- Large areas tend to look more like they do on **the globe** than is true for other projections.

Conic projections- Tangential vs Secant



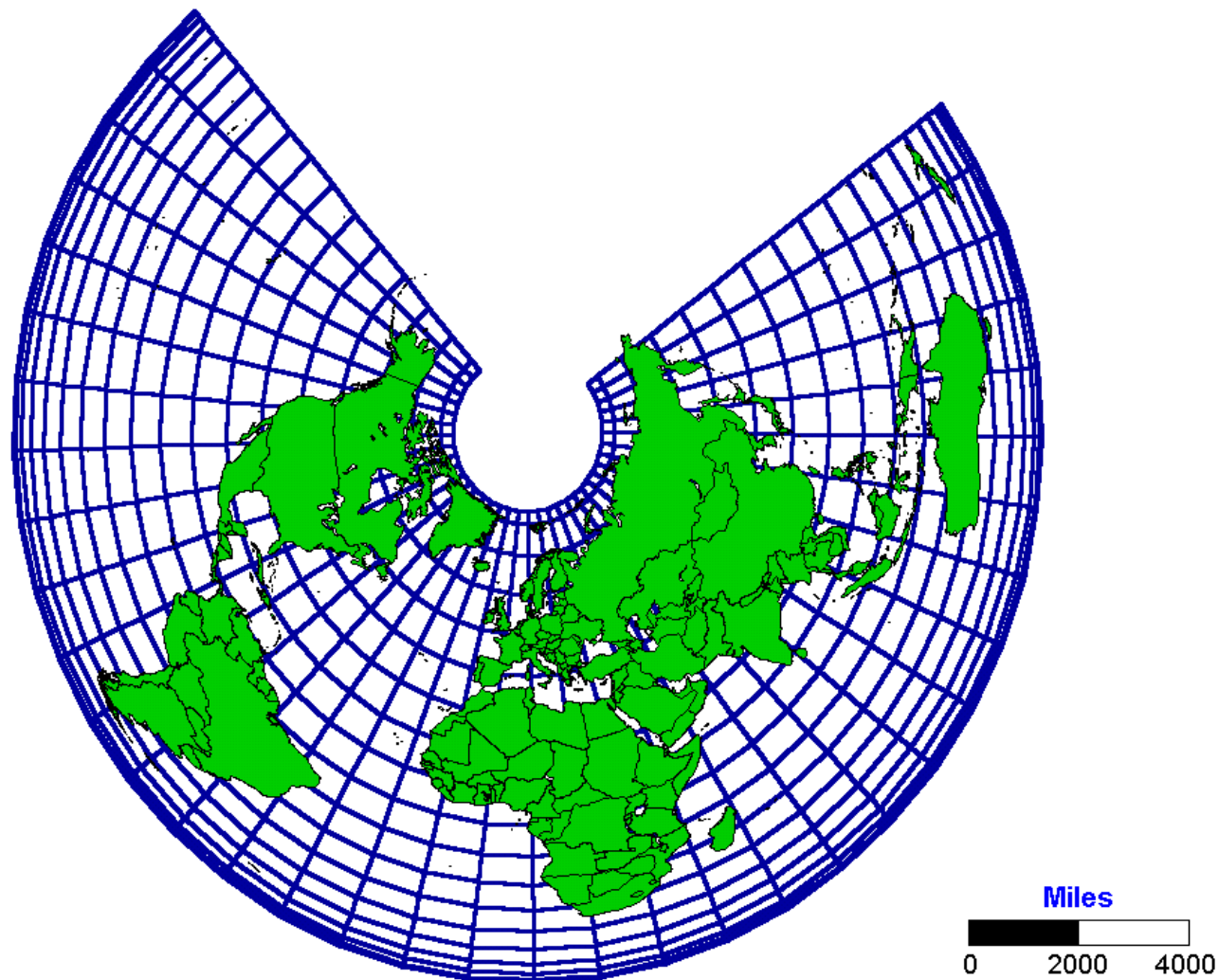
In tangential **one line** is true where others are distorted;
In secant 2 lines are true

Conic projections- Secant

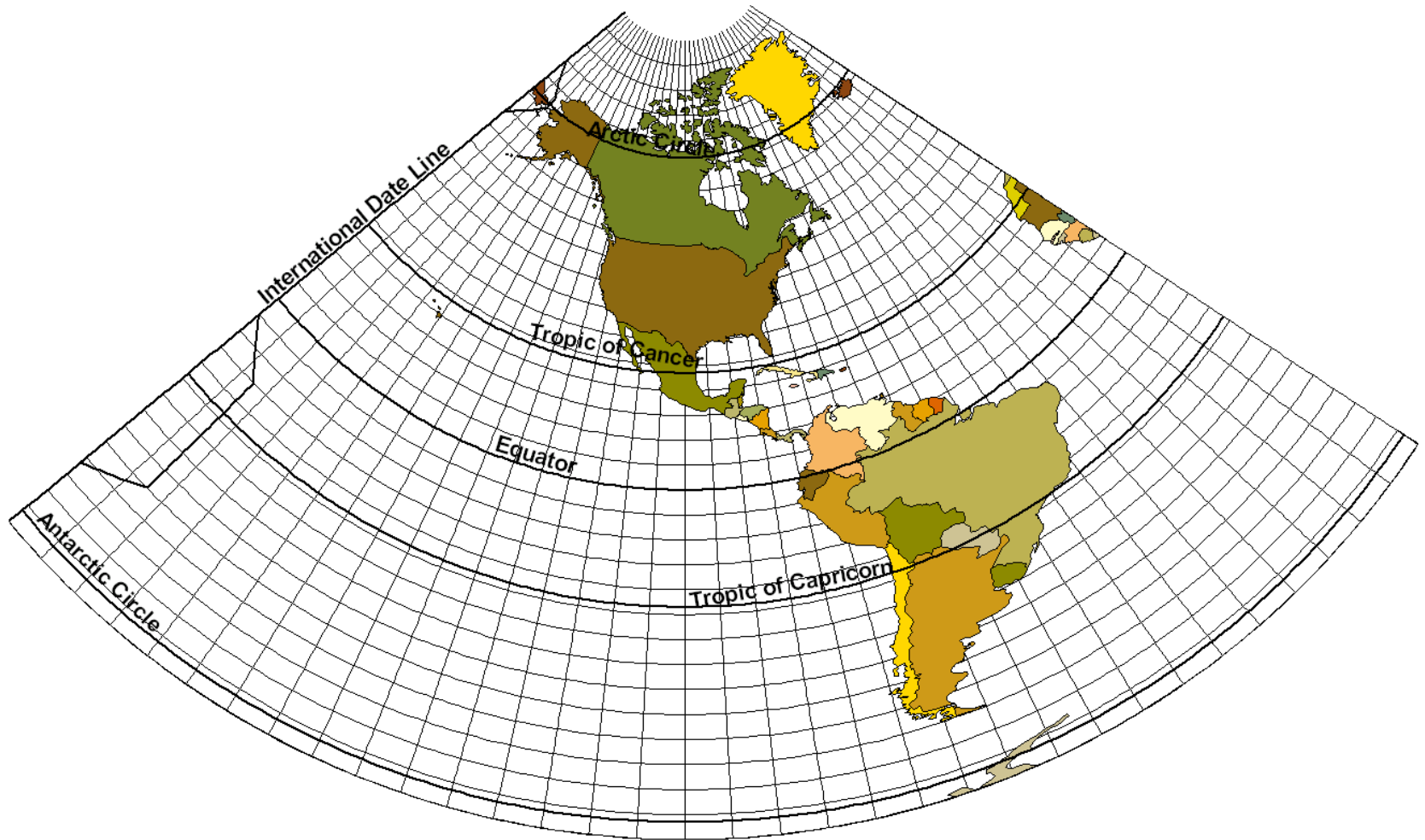


- Globe projected onto a cone, which is then flattened.
- Cone usually fit over pole like a dunce cap.
 - Meridians are **straight lines**.
 - Angle between all meridians is **identical**
- Usually drawn secant (**converse area**).
- Area between standard parallels is “projected” inward to cone (**converse area**).
- Areas outside standard parallels projected outward.

Albers Conic Projection



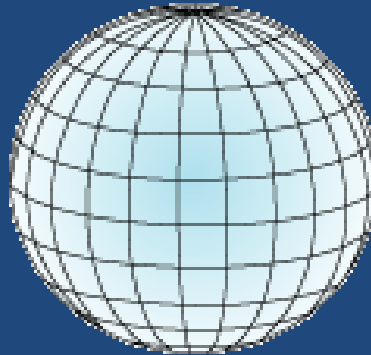
Equidistant Conic Projection



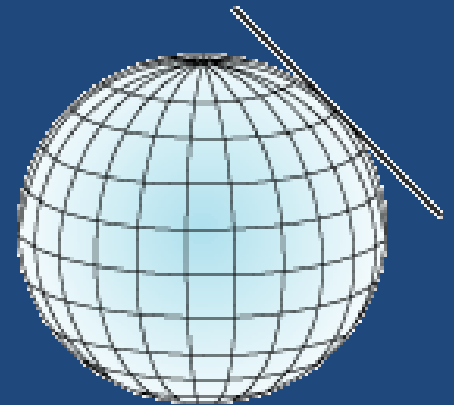
Azimuthal (also called stereographic or orthographic) Projections



Polar



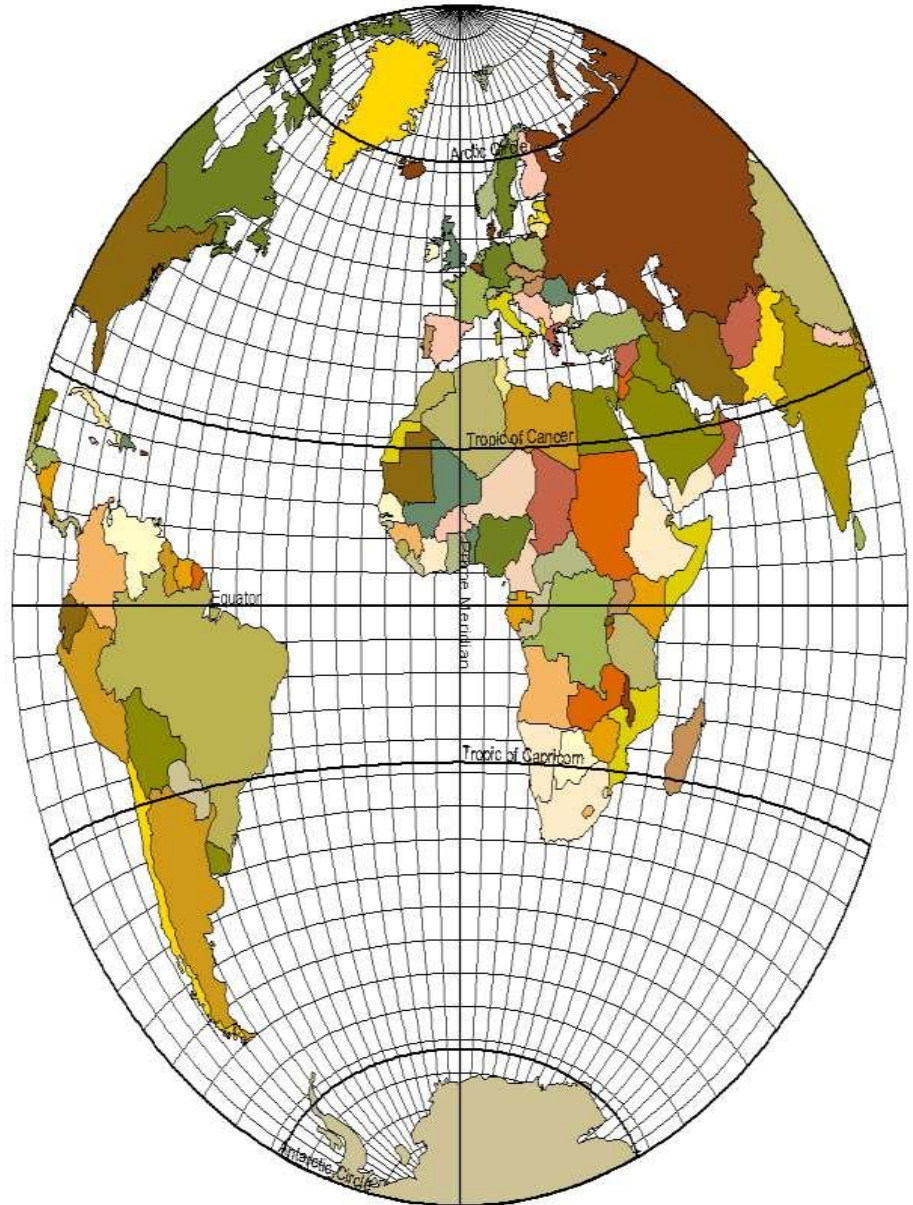
Equatorial



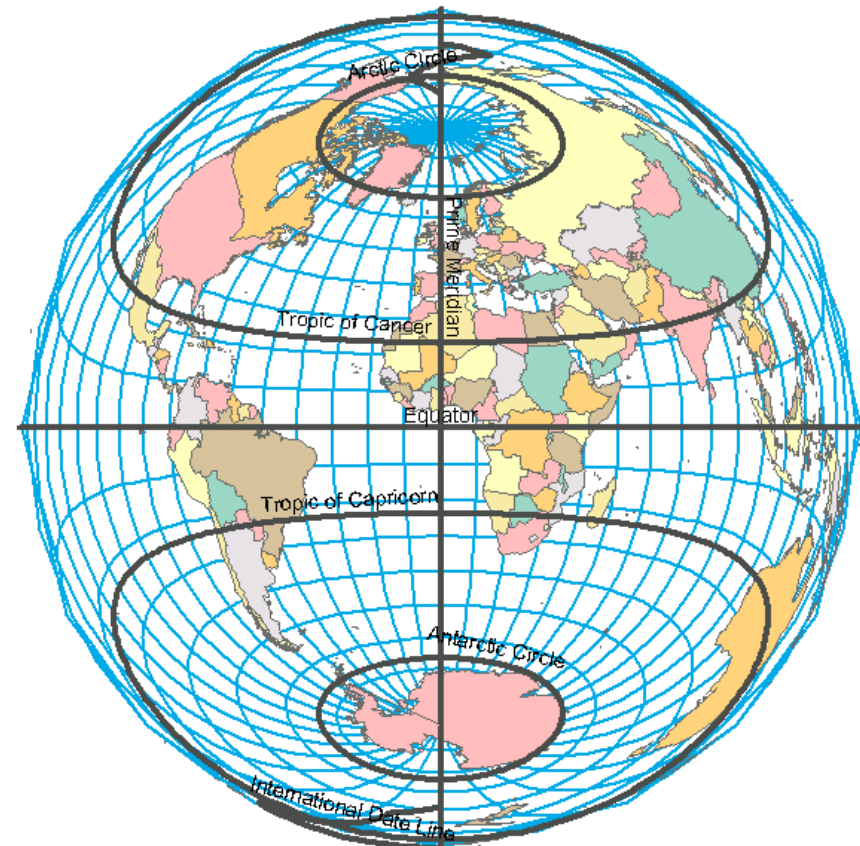
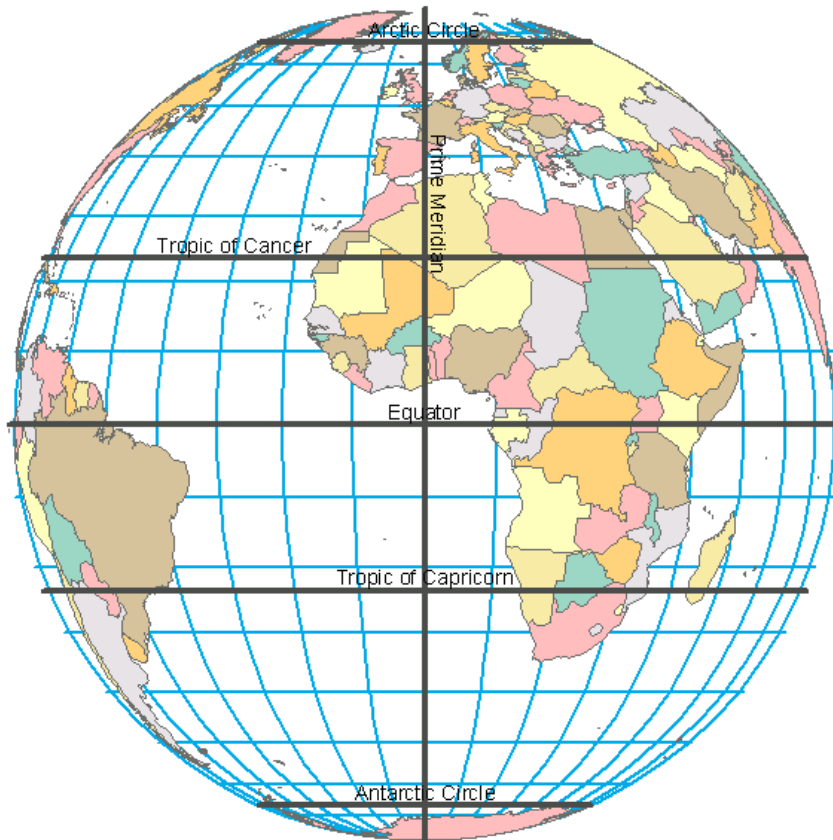
Oblique

Planar projection

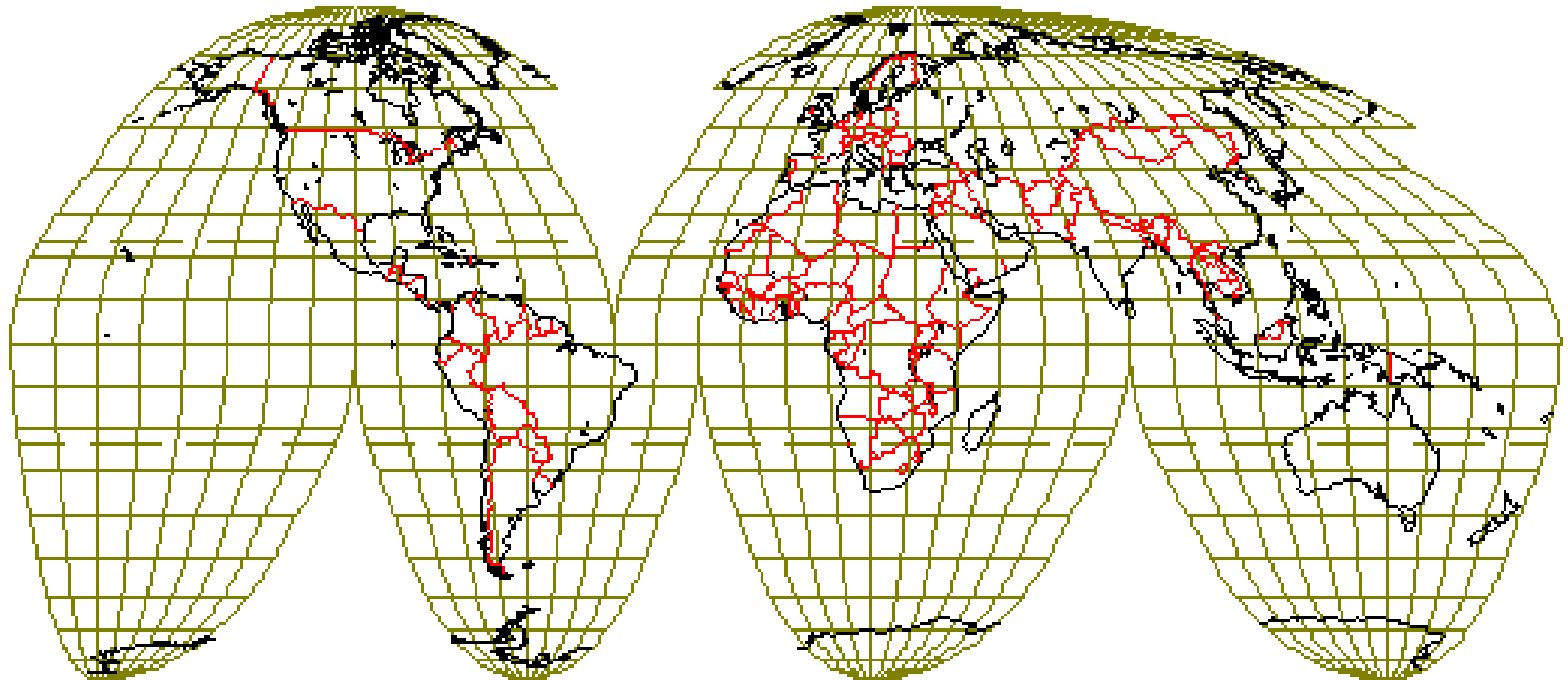
- Earth grid and features projected from sphere to a **plane surface**
- **Length** of a straight line between two points represents correct great **circle distance (Conserve distance)**.
- Lines to measure distance can *originate at only one or two points*



Plane Projection: Lambert Azimuthal Equal Area



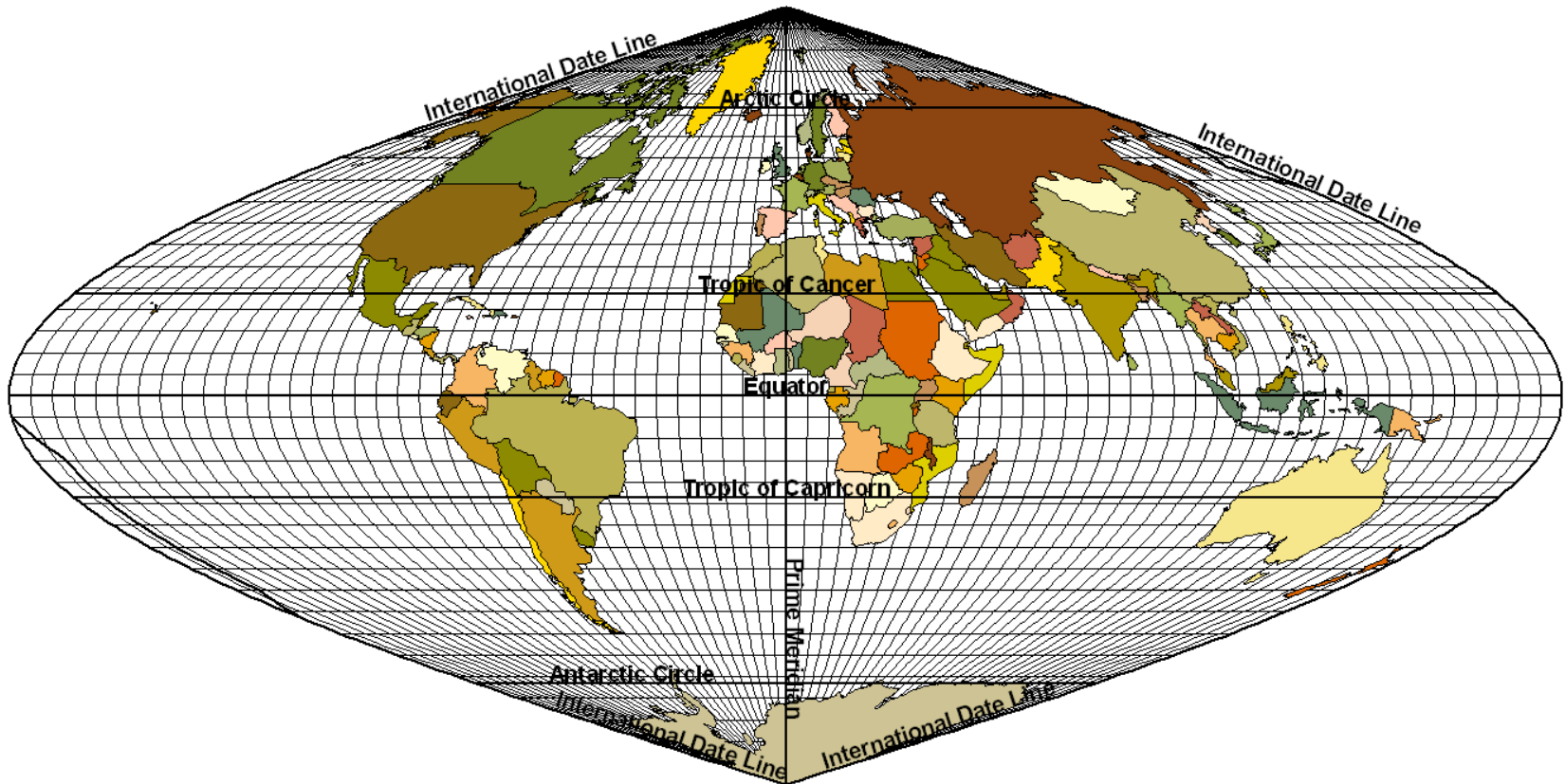
Special projections



GOODE'S HOMOLOGOSINE EQUAL AREA PROJECTION

MicroCAM

Sinusoidal Equal Area Projection



Buckminster Fuller's "Dymaxion"



Characteristics of projections

- Regardless of what type of map you make (using whatever projection you choose) the representation of the Earth will **be distorted** in at least 1 (or more) aspects

Shape- maps that preserve shape are called conformal maps

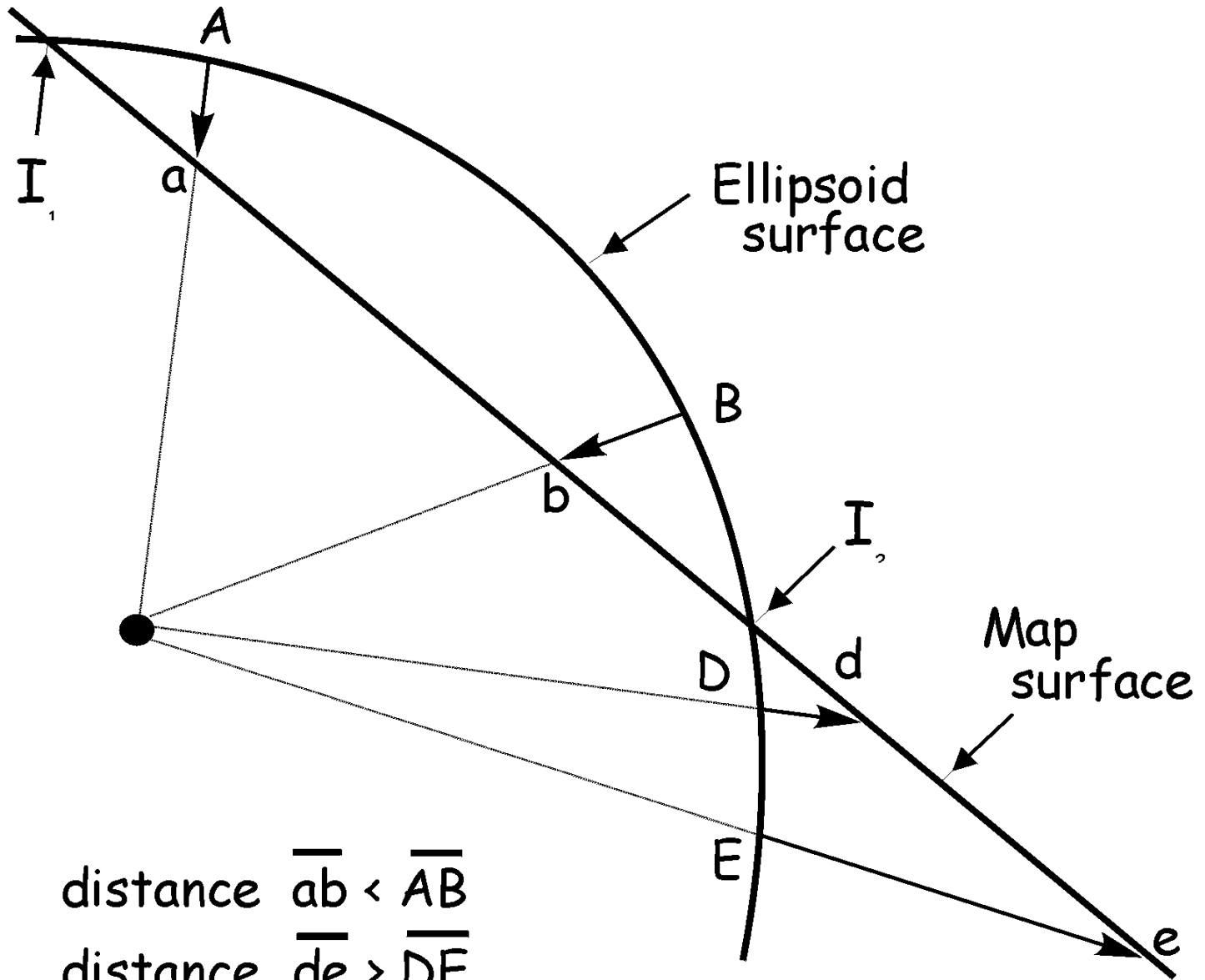
Area- equal area maps preserve the actual amount of the surface area the region takes up on the globe relative to other areas of interest

Distance- equidistant maps preserve distance for all lines that pass through a specified point

Direction- azimuthal maps preserve direction for at least one point on the map, such that all lines that pass through that point show true direction.

- **No map** can show all 4 things without distortion, and no map is able to preserve all of the characteristics equally, regardless of the type

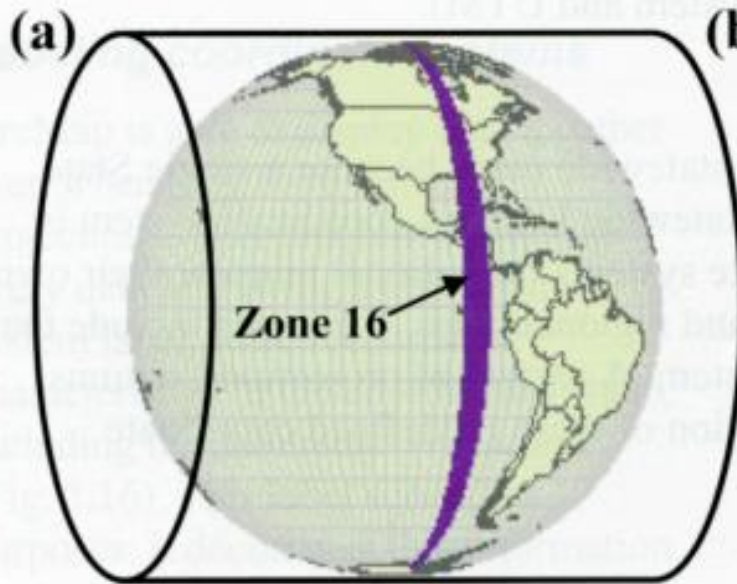
- In any projection there is no distortion on the line where the globe touches the cone, plane or cylinder (i.e., there's always at least one true set of coordinate lines)



Popular Projection

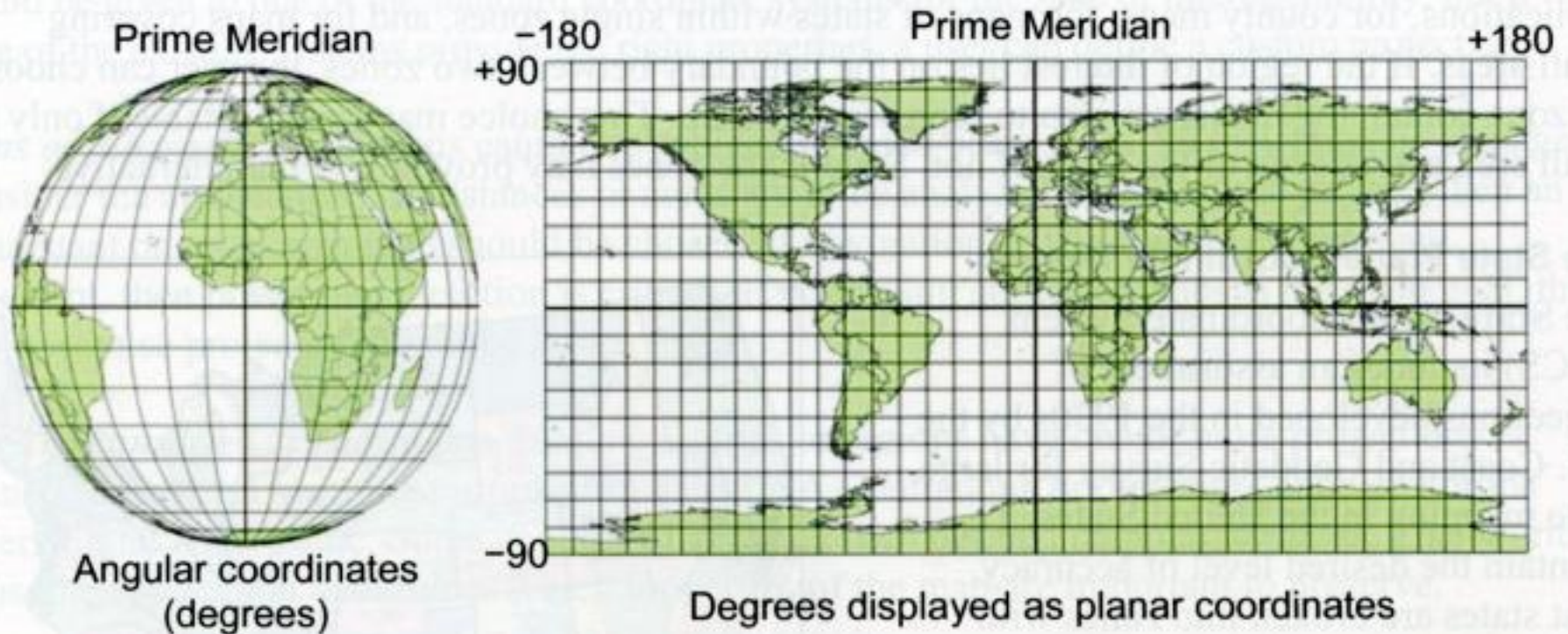
Universal Transverse Mercator (UTM)

Based on Secant transverse cylindrical projection Divides up Earth's surface into zones that are approximately 180 KM across (where the secant of the cylinder occurs as it intersects the globe's surface).



Popular Projection

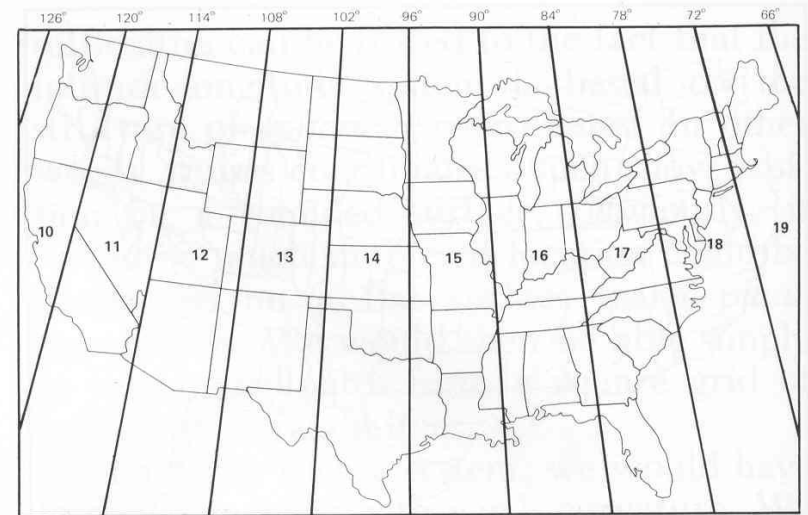
Universal Transverse Mercator (UTM)



It is widely used because direction is not distorted and can be used for navigation. Area and shape are distorted (Look how big Greenland appears compared to Australia).

Universal Transverse Mercator system

- Based on a cylindrical projection running from pole-pole
- Distortion minimized in a N – S “strip” (zone)
- Zones are 8° wide but overlap by 1° on each side. 60 world wide.



7.7 Zones of the Universal Transverse Mercator Grid in the contiguous United States.

UTM coordinates

- Coordinates are based on an arbitrary origin at equator and 500,000 m west of central meridian
- E-W position: “easting”
N-S position: “northing”

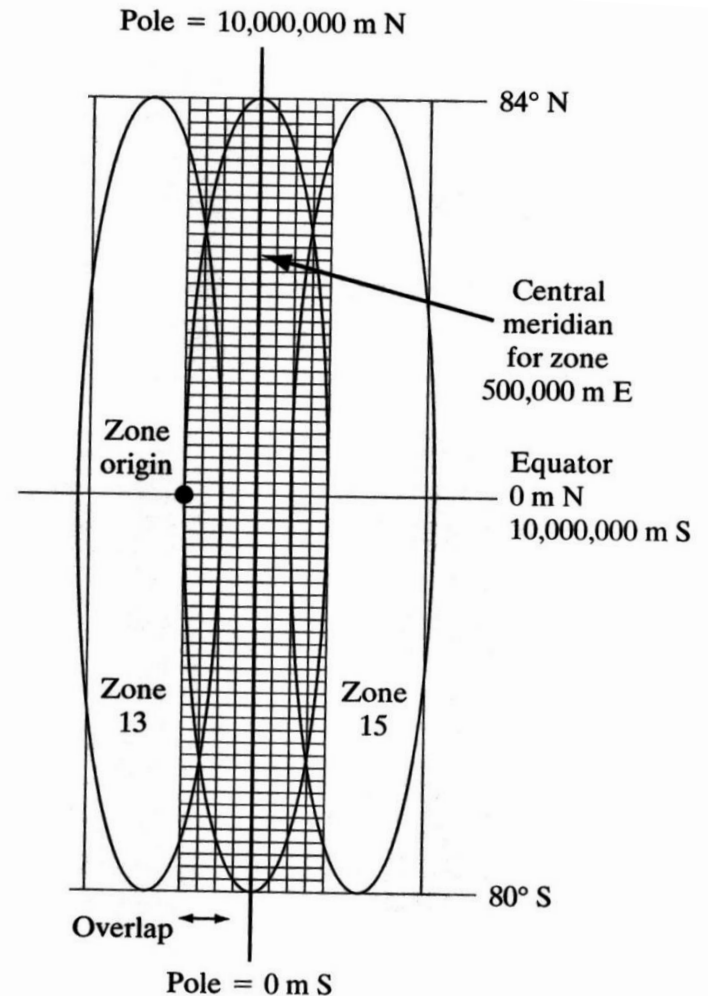


FIGURE 2.25
The Universal Transverse Mercator (UTM) coordinate system. (Source: Clarke, 1999)

Miscellaneous Coordinate Systems

- Military grids
- Land survey grids
- Cadastral records
- Other ...

Remarks: the cartographic process

1. Model surface of Earth mathematically
2. Create a geographical datum
3. Project curved surface onto a flat plane
4. Assign a coordinate reference system

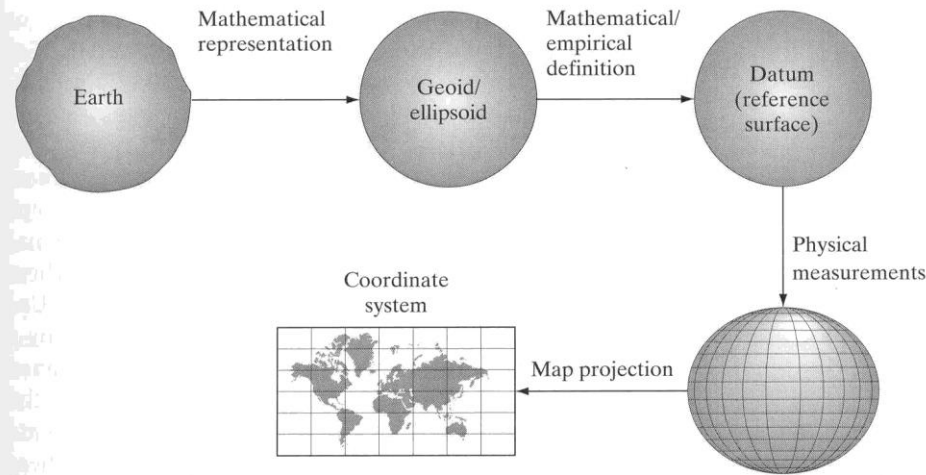


FIGURE 2.14

The concept of georeferencing. Georeferencing involves a series of transformations that progressively flattens the irregular surface of Earth so that measurements on the curved surface on Earth can be represented on a flat surface of a map.

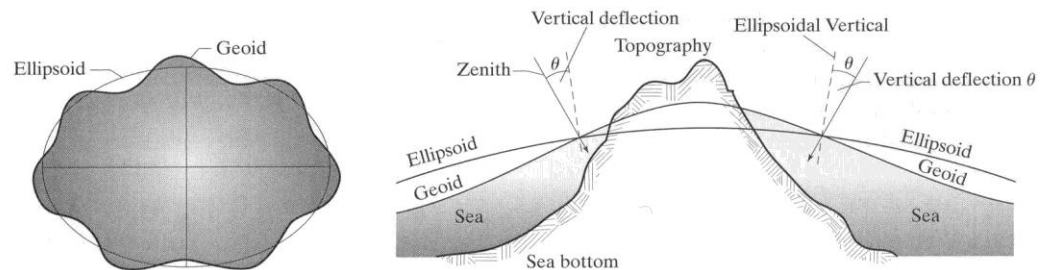


FIGURE 2.15

The relationships between Earth's irregular surface, ellipsoid, and geoid. (Source: Seeber, 1993)

Geoids: vertical reference surfaces

- Like MSL (mean sea level) extended across continents
- Based on network of precise gravity measurements
- Can depart from ellipsoid by as much as 60 m

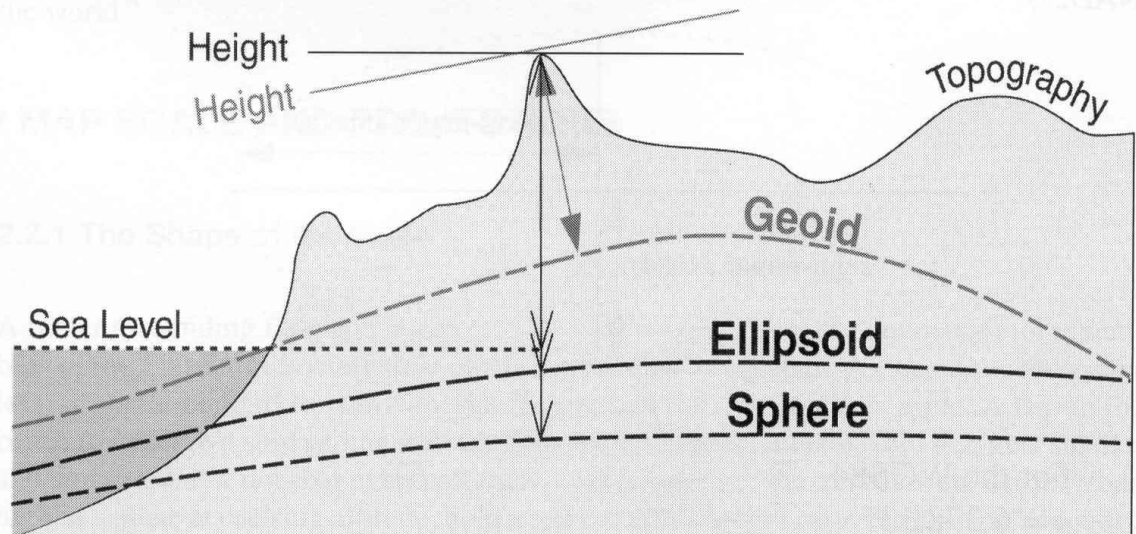


Figure 2.4 Elevations defined with reference to a sphere, ellipsoid, geoid, or local sea level will all be different. Even locations as latitude and longitude will vary somewhat. When linking field data such as GPS with a GIS, the user must know what base to use.

2. Land surveying

(Material of Sz. Rózsa: Surveying I)

Surveying:

The art of making measurements of the **relative positions** of natural and man-made features on the Earth's surface, and the presentation of this information either graphically or numerically.

Geodesy:

Geodesy is the discipline that deals with the measurements and representation of the Earth, including its gravity field, in a three-dimensional time varying space.

Geodesy focus on the Earth and neglect any man-made features on it (e.g. buildings, public utilities, etc.), while surveying use the results of geodesy for positioning and mapping of these features.

Basic principles of Surveying

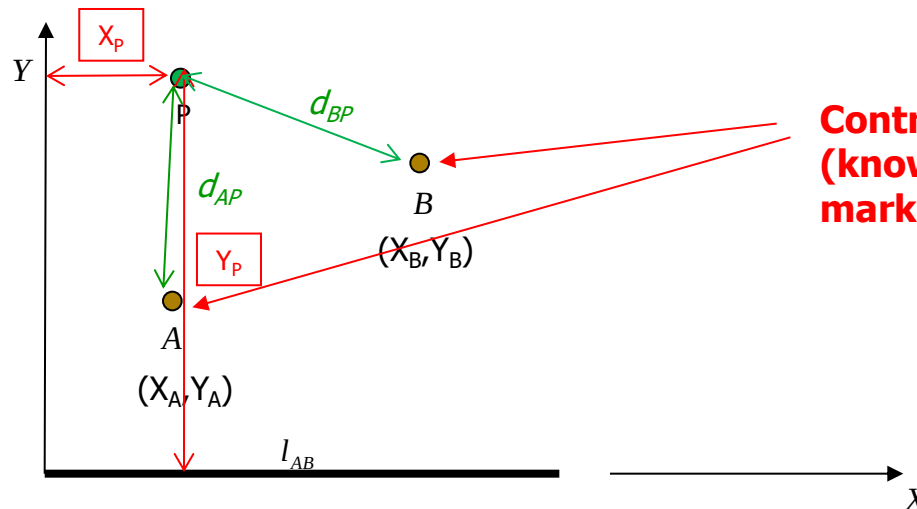
Recall the definition of Surveying:

The art of **making measurements of the relative positions** of natural and man-made features on the Earth's surface, and the **presentation of this information** either graphically or numerically.

How to achieve this?

Let's determine the position (X_P , Y_P) of point P!

Absolute vs **Relative** positioning



The positioning is usually separated into horizontal (2D) and vertical (1D) positioning.

Nowadays 3D positioning can be achieved using satellite techniques, too.

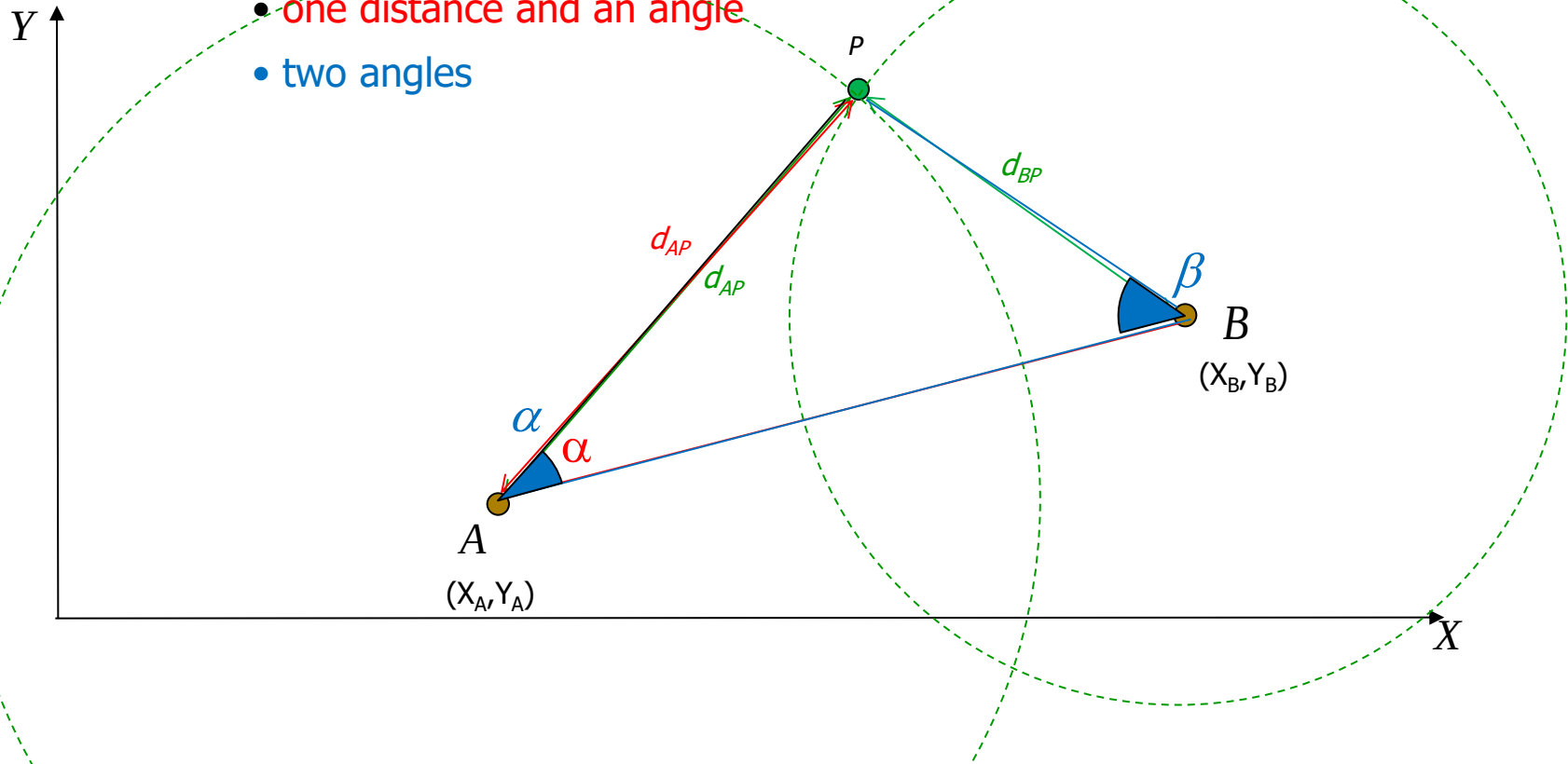
Basic principles of Surveying

Let's determine the position of a third, unknown point (P).

We have two unknowns: X_P , Y_P

We need two measurements:

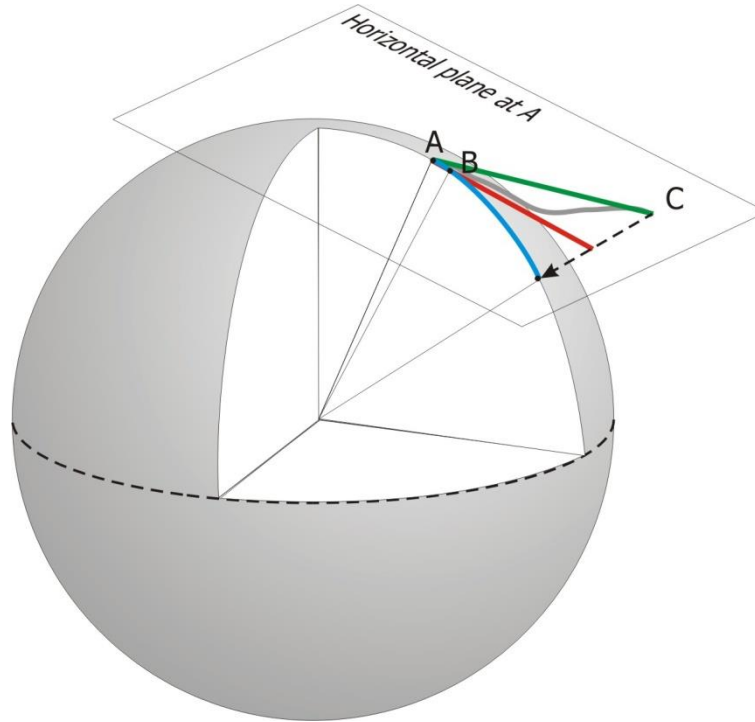
- two distances
- one distance and an angle
- two angles



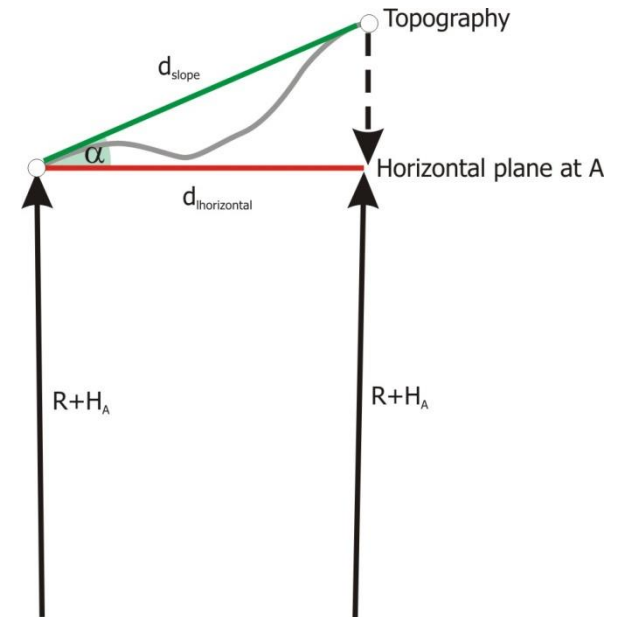
Classification of Surveying

According to the space involved:

Plane Surveying

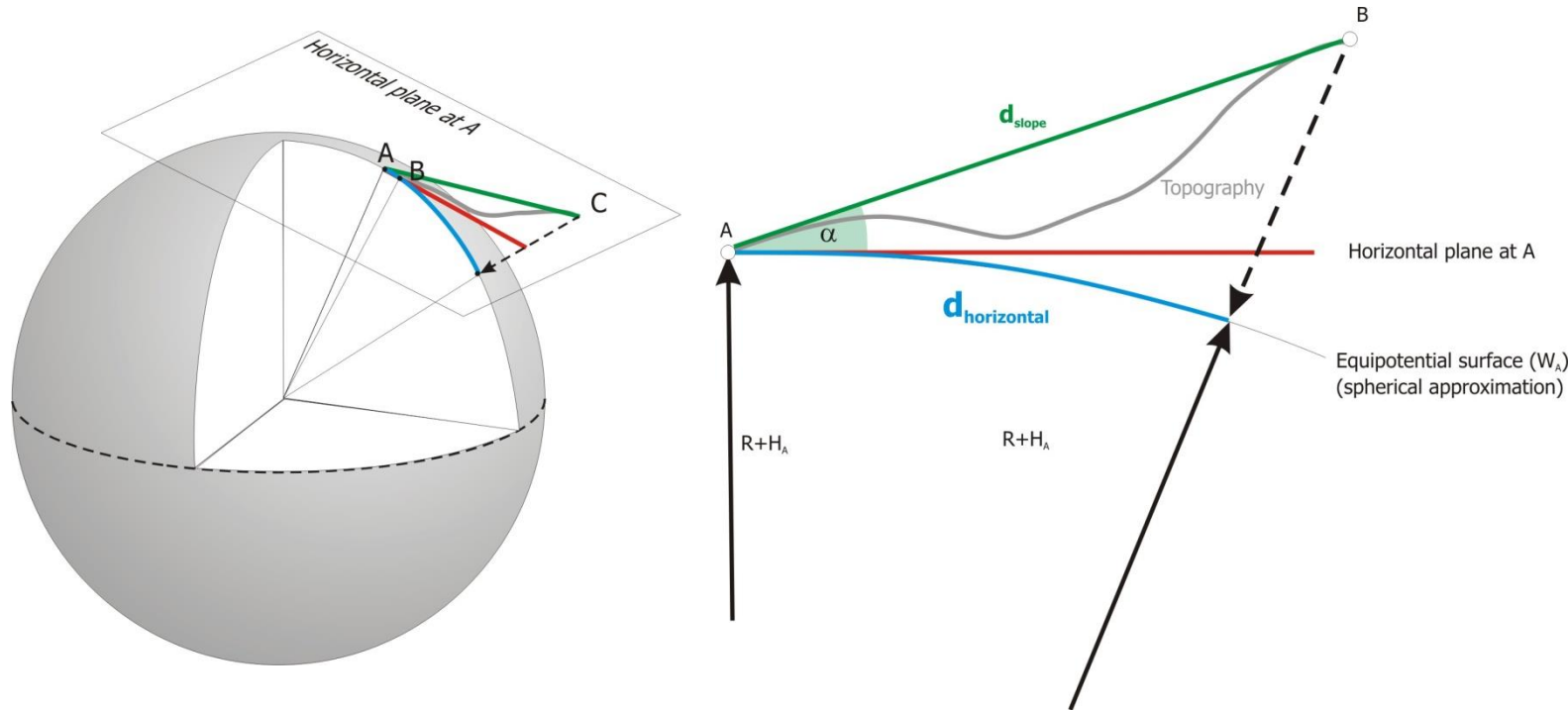


- relatively small areas
- surface of earth can supposed to be flat
- measurements plotted represent a horizontal projection of the actual field measurements



Note: The two radii can supposed to be parallel, when the $l(A,B)$ is small.

Classification of Surveying



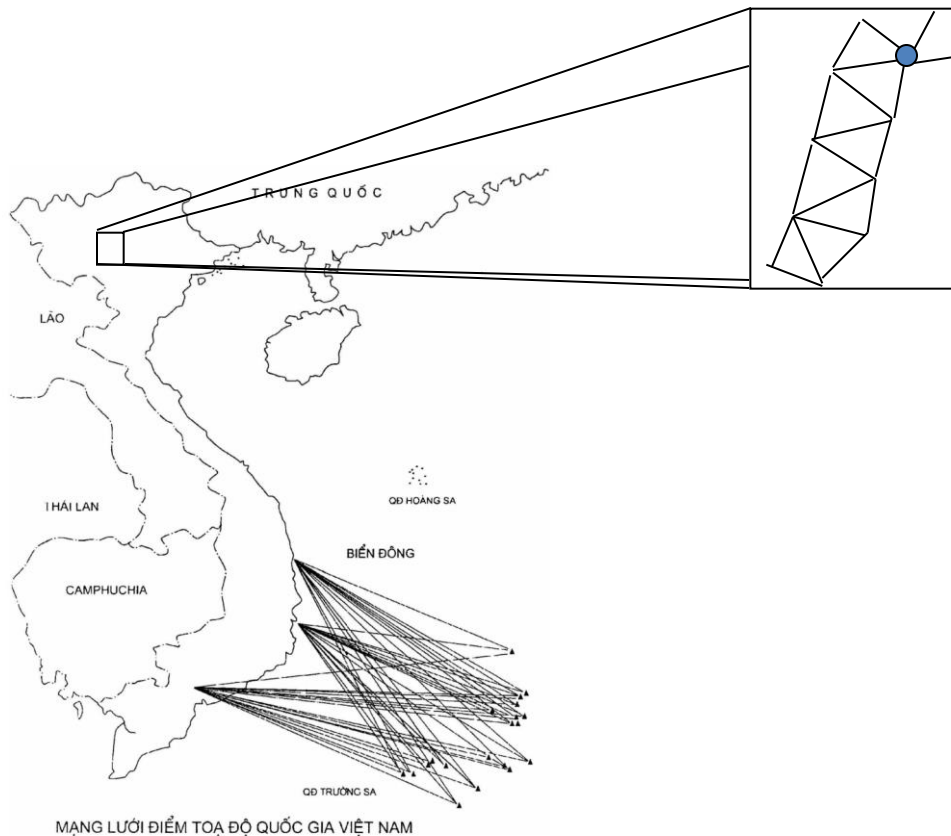
- large areas
- surface of the Earth can not supposed to be flat
- the curvature of the Earth is taken into account

Mostly used for establishing control networks, determining the size and shape of the Earth and determining the gravity field of the Earth.

How to create a countrywide coordinate system?

In order to use the relative positioning, a proper number of control points are needed. These points:

- are coordinated points;
- are marked.



Control Networks

Why is it necessary to have a common countrywide coordinate system?

Many engineering tasks cover a large area (highways, bridges, tunnels, channels, land registry, etc.), where the common coordinate system (reference system) should be available.

The Control Network provide us with control points given in the same reference system (coordinate system).

Thus measuring the relative positions of unknown points using these control points, the coordinates of the new points can be computed in the same reference system.



The role of Surveying in Civil Engineering Practice

Surveyors are needed:

- to maintain the geometric order during the construction process



What is this?



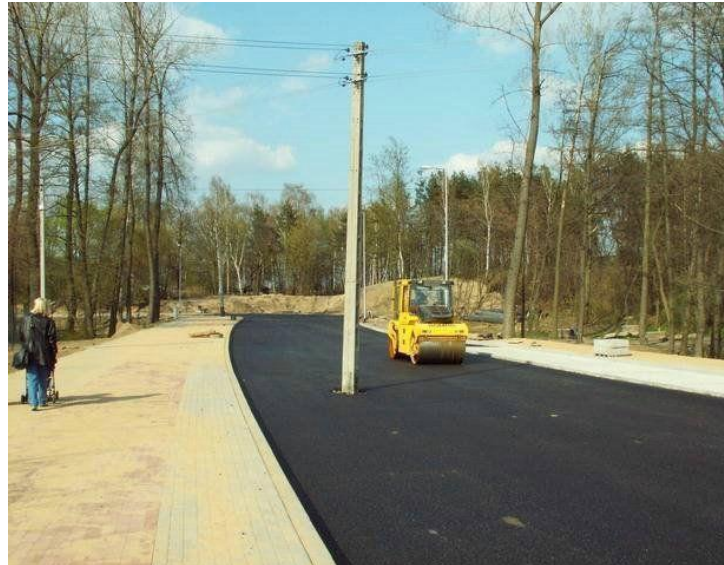
Laying them in the appropriate geometry, outstanding structures can be created!

- to provide fundamental data for the design and planning process

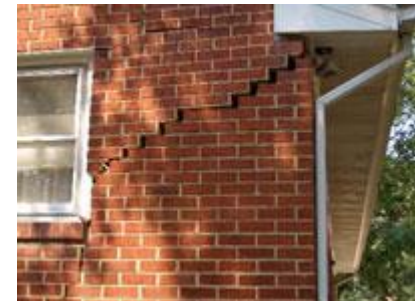


Wrong geometry – the structure is not functional!

- to provide quantity control during the construction process (for example: earthwork quantities)

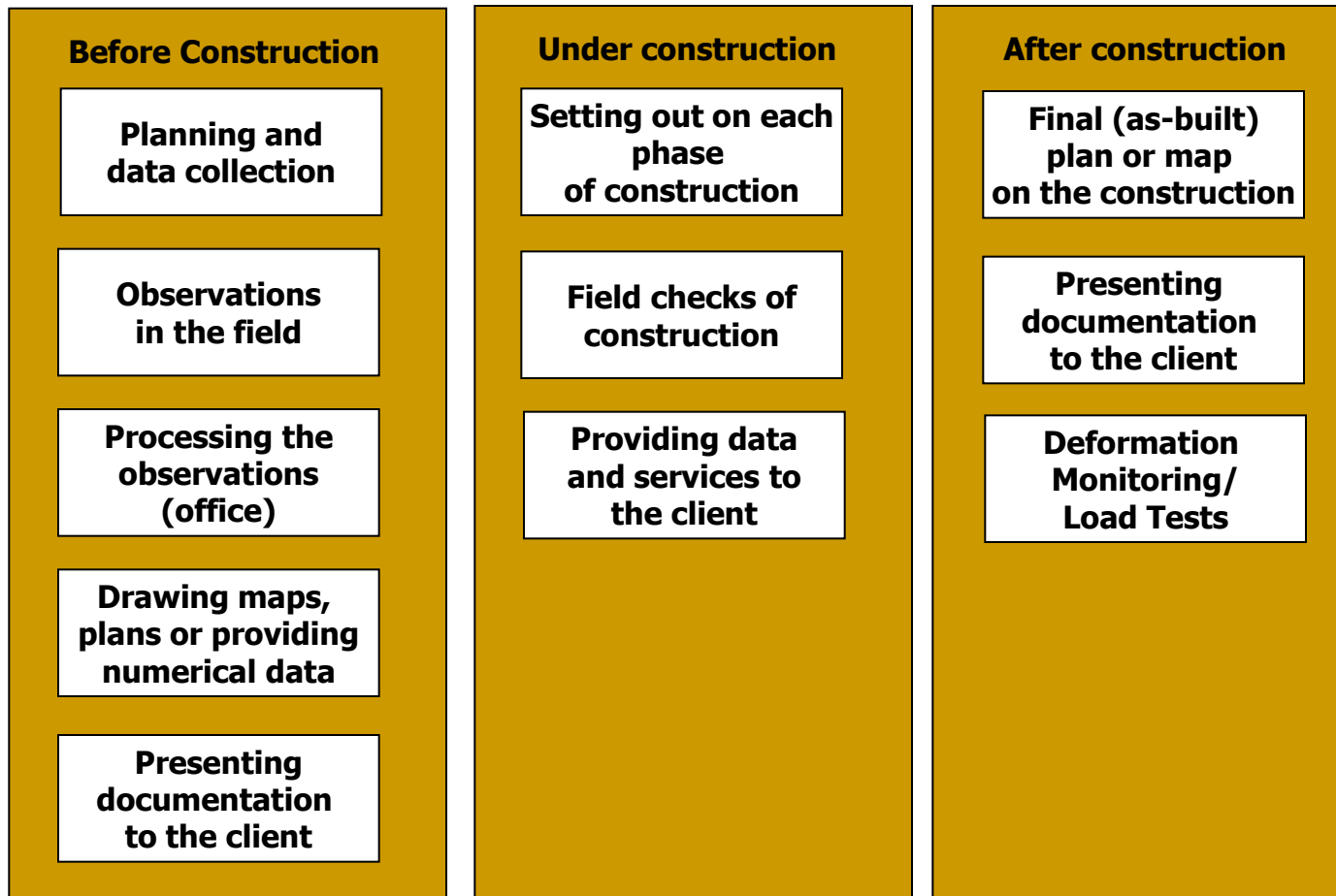


- to monitor the structure after the construction (subsidence, deformations, etc.)



The role of Surveying in Civil Engineering Practice

Surveying activities during the construction process



Methods of height determination

Question 1:

What does the height (elevation) of a point mean?

Question 2:

What does it mean, when point B is at a higher elevation than point A?

Answer 1:

The height of a point represents its energy level above a reference level.

Answer 2:

For example water flows from point B to point A.

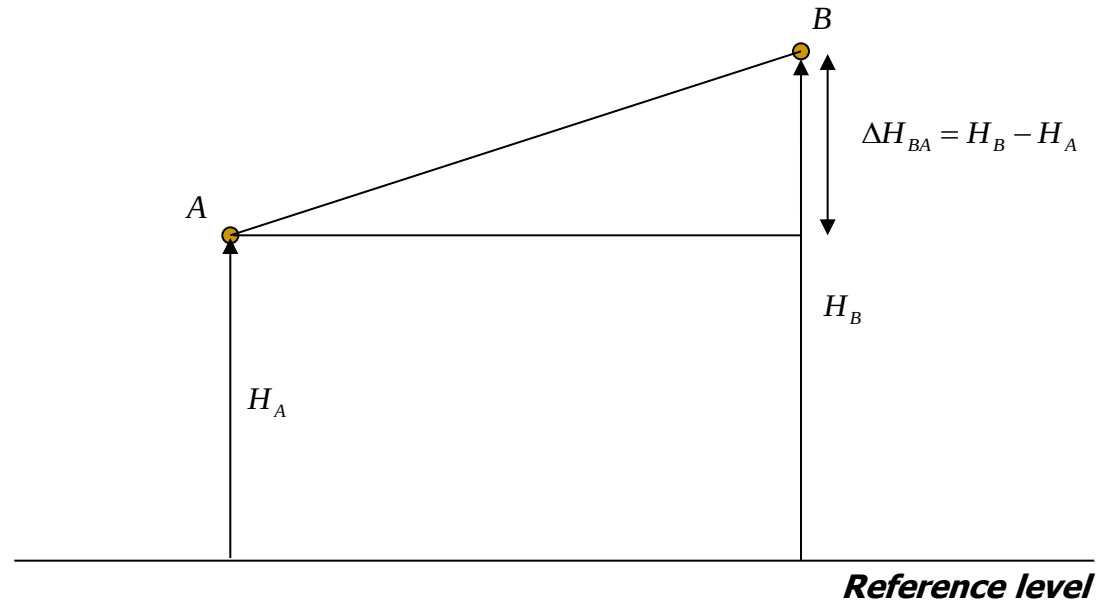
Methods of height determination

1D position determination - determining the height

We can not determine absolute heights above the reference level

Relative height determination - determining the height differences

Levelling benchmarks are needed - control points for which the elevation is known.

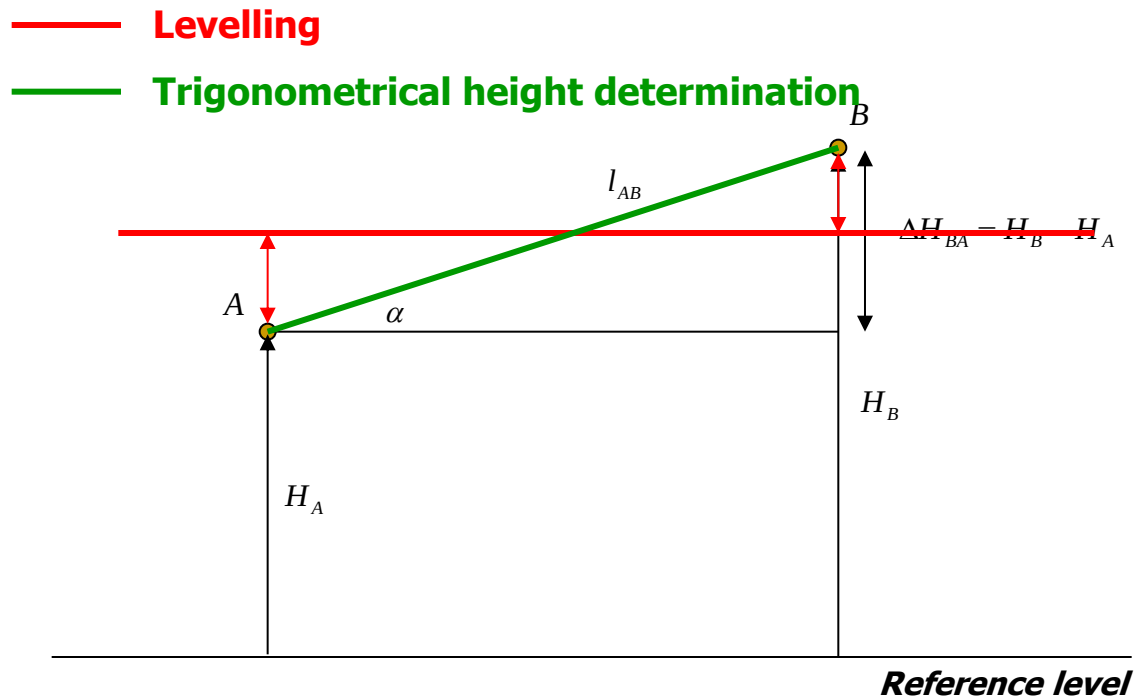


Methods of height determination

How can we determine the height difference?

Two solutions:

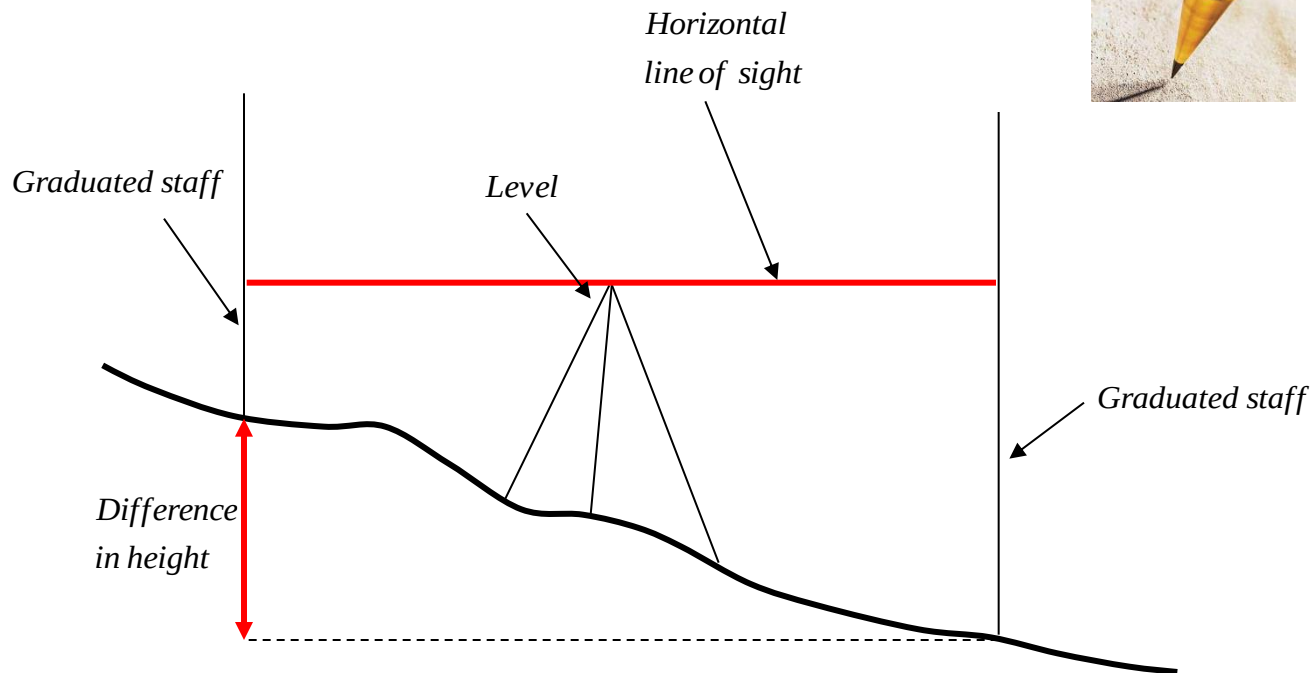
- setting a horizontal plane, and measuring the offset from this plane
- measuring the slope and slope distance between the points

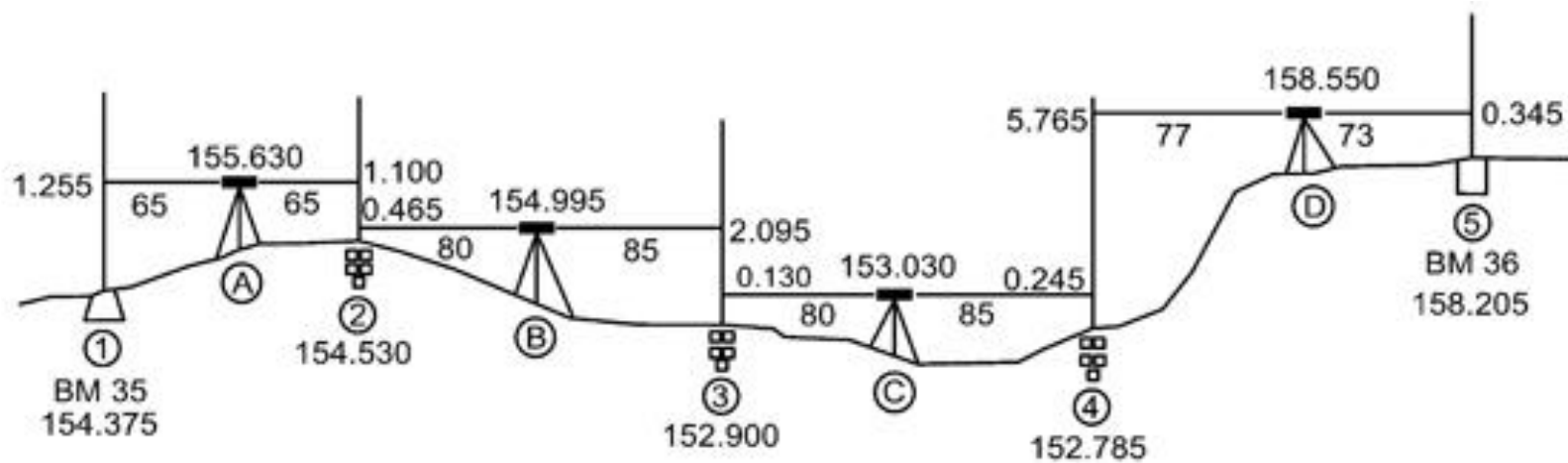


Levelling

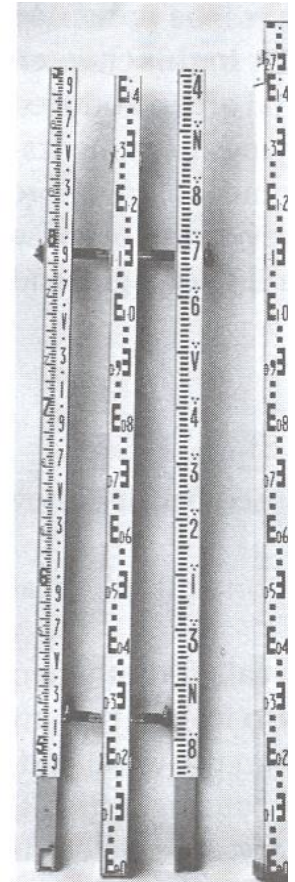
Over short distances the horizontal line and level line coincide.
For a distance of 100m the effect of the curvature is less than 1 mm.

The levelling device (called level) must be set up so, that the line of sight is perpendicular to the gravity vector (plumb line). -> the line of sight is horizontal.





Levelling

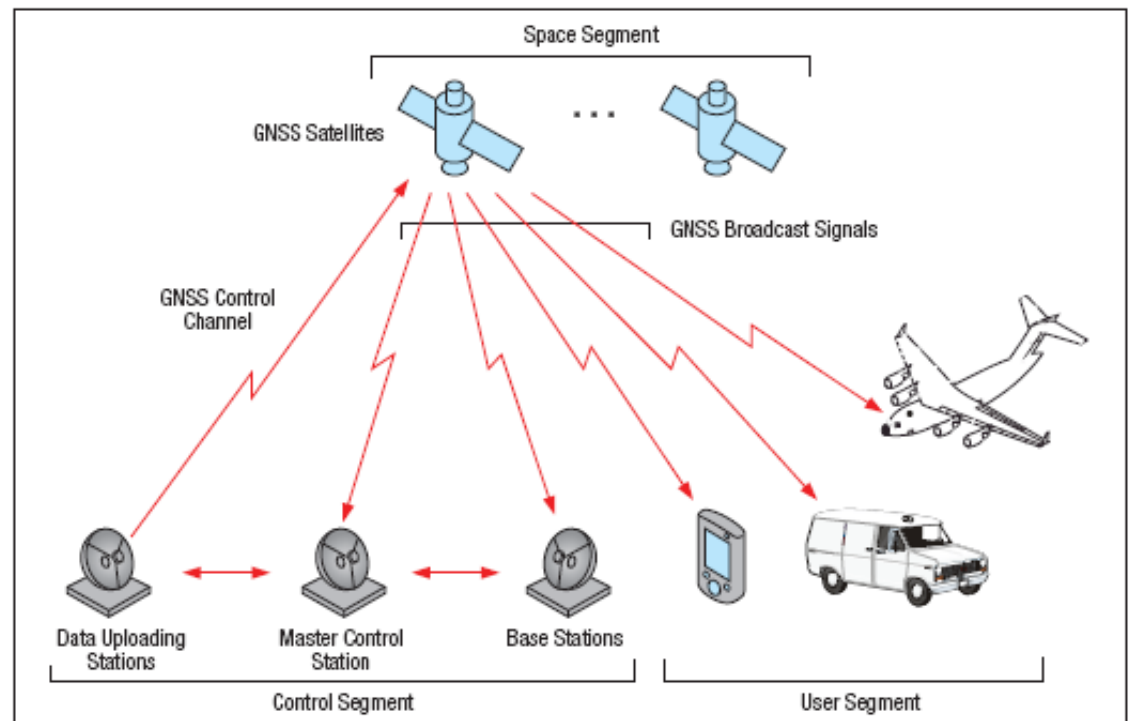


3. Global navigation satellite systems (GNSS)

- GNSS (Global Navigation Satellite Systems) started with the launch of the U.S Department of Defense Global Positioning System (GPS) in the late 1970's
- GNSS systems currently include
 - GPS (United States)
 - GLONASS (Russia)
 - Galileo (European Union)
 - BeiDou (China)

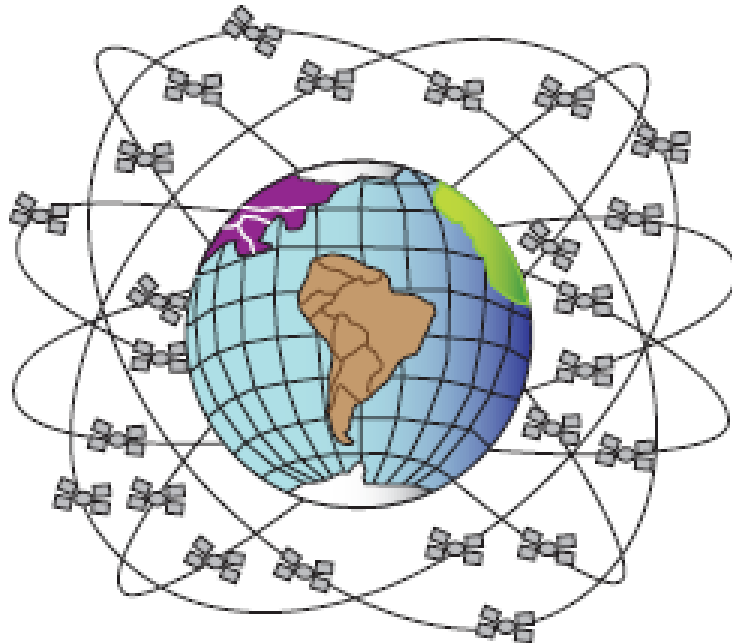
Architecture

- GNSS satellite systems consists of three major components or “segments”:
 - Space Segment
 - Control Segment
 - User Segment



Space Segment

- Consists of GNSS satellites, orbiting about 20,000 km above the earth. Each GNSS has its own constellation of satellites

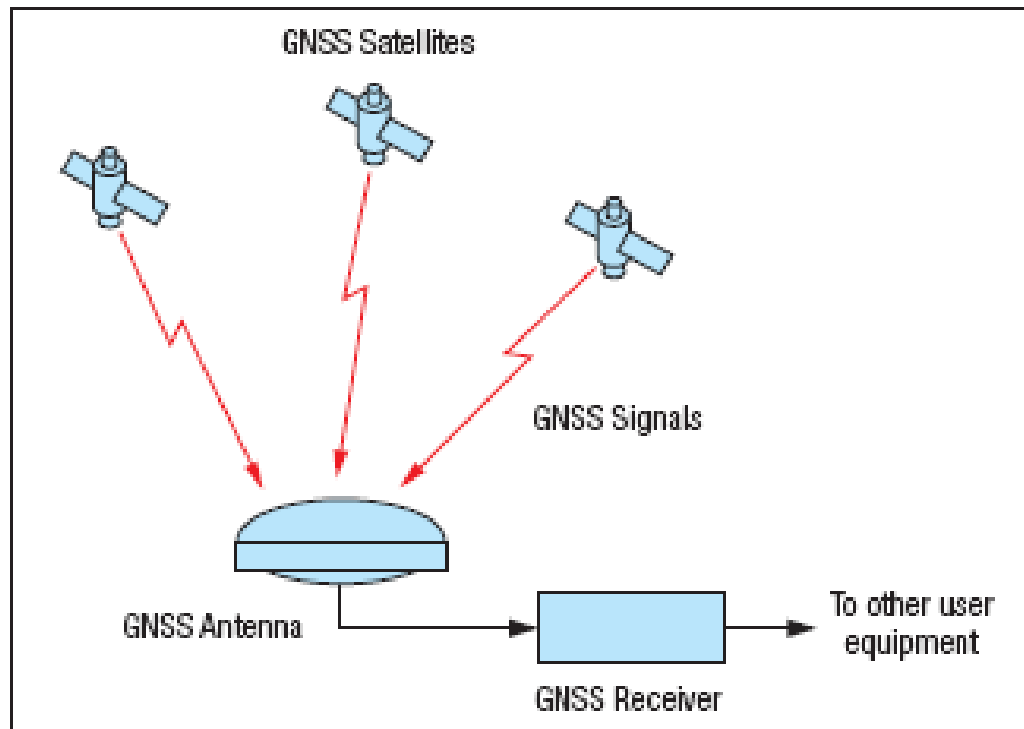


Control Segment

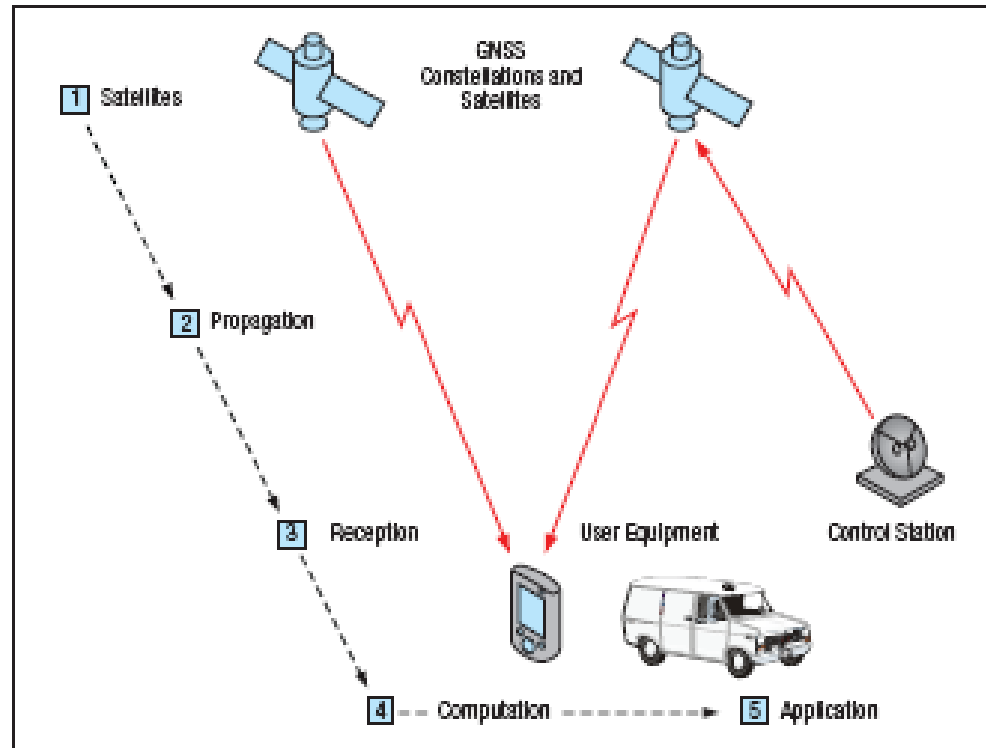
- The control segment comprises of a **ground-based network** of master control stations, data uploading stations, and monitor stations.
- Master control stations **adjust** the satellites' orbit parameters and on-board high-precision clocks when necessary to maintain accuracy
- Monitor stations **monitor** the satellites' signal and status, and relay this information to the master control station
- Uploading stations **uploads** any change in satellite status back to the satellites

User Segment

- User segment consists of GNSS antennas and receivers used to determine information such as position, velocity, and time



Basic GNSS Concepts



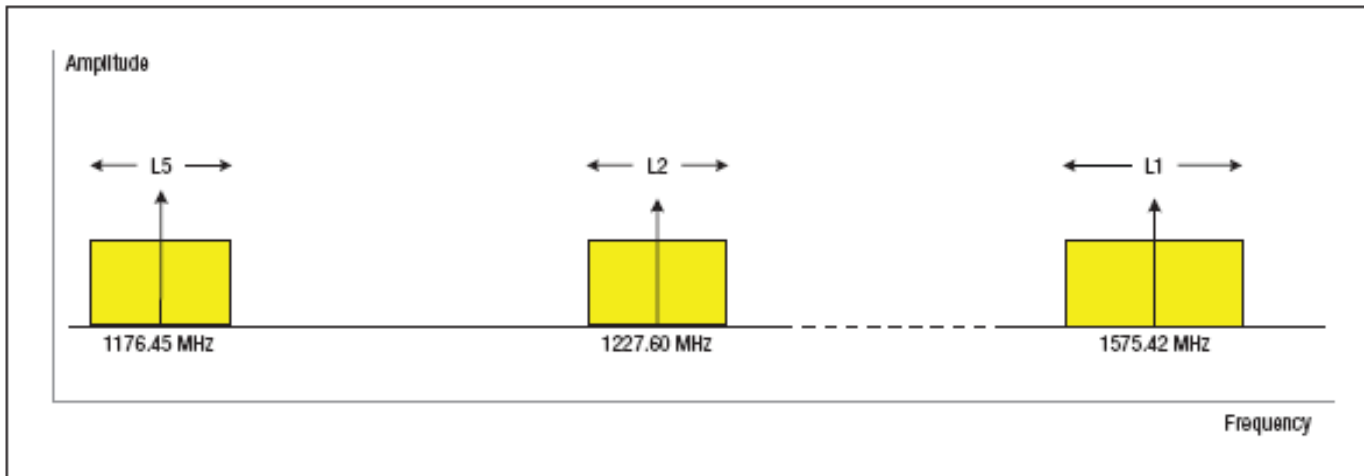
The above figure shows the steps involved in using GNSS to determine time and position then applying the information.

Satellites

- Multiple GNSS constellations orbiting the earth
 - Beneficial in difficult environment with obstructions to direct line of sight to satellites.
Multiple constellations will give more observations
- GNSS satellites know their time and orbit ephemerides very accurately
- Timing accuracy is very important. The time it takes a GNSS signal to travel from satellites to receiver is used to determine distances (range) to satellites
- 1 microsecond = 300m, 1 nanosecond = 30 cm.
- Small deviations in time can result in large position errors

Satellites

- GPS transmits at the following frequencies



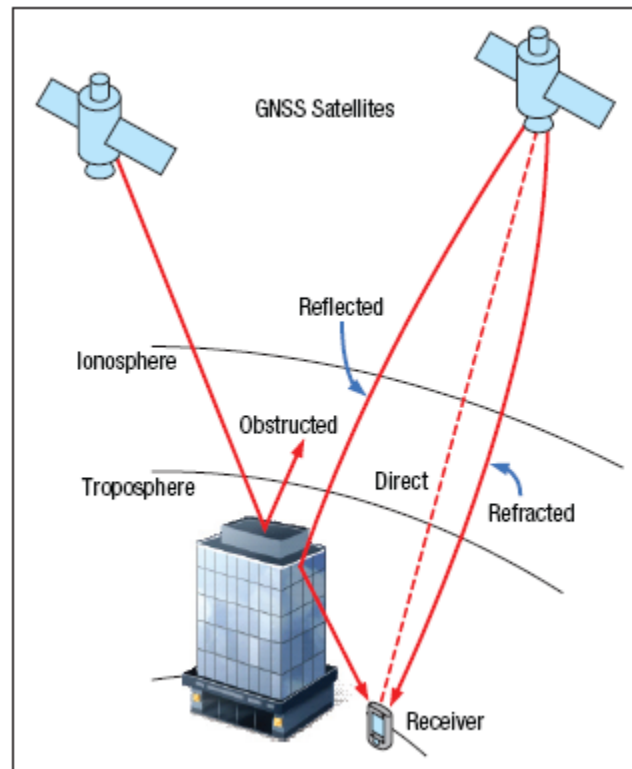
- This frequency band is referred to as the L-band, a portion of the radio spectrum between 1 and 2 GHz
- L1 transmits a navigation message,
Coarse acquisition (C/A) code is freely available to public
Encrypted precision (P) code, called the P(Y) code (restricted access), is transmitted on both L1 and L2.

Satellites

- Navigation message includes the following information:
 - GPS date and time
 - Satellite status and health
 - Satellite ephemeris data, which allows the receiver to calculate the satellite's position.
 - Almanac, which contains information and status for all GPS satellites
- The P(Y) code is for military use, and provides better interference rejection than the C/A code.
- Newer GPS satellites now transmits L2 C/A code (L2C), providing a second publicly available code to civilian users.

Propagation

- GNSS signals pass through the near-vacuum of space, through the various layers of the atmosphere to the earth

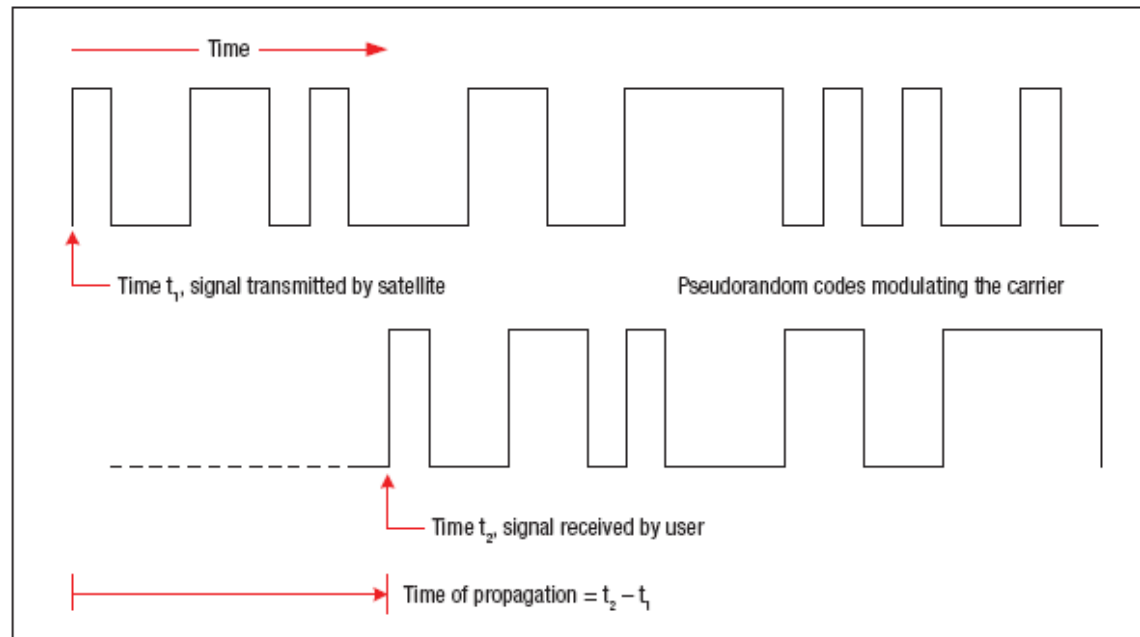


Reception

- Receivers need at least 4 satellites to obtain a position. If more are available, these additional observations can be used to improve the position solution
- GNSS signals are modulated by a unique pseudorandom digital sequence, or code. Each satellite uses a different pseudorandom code
- Pseudorandom means that the signal appears random, but actually repeats itself after a period of time
- Receivers know the pseudorandom code for each satellite. This allows receivers to correlate (synchronize) with the GNSS signal to a particular satellite
- Through code correlation, the receiver is able to recover the signal and the information they contain

Reception

- For each satellite tracked, the receiver determines the propagation time

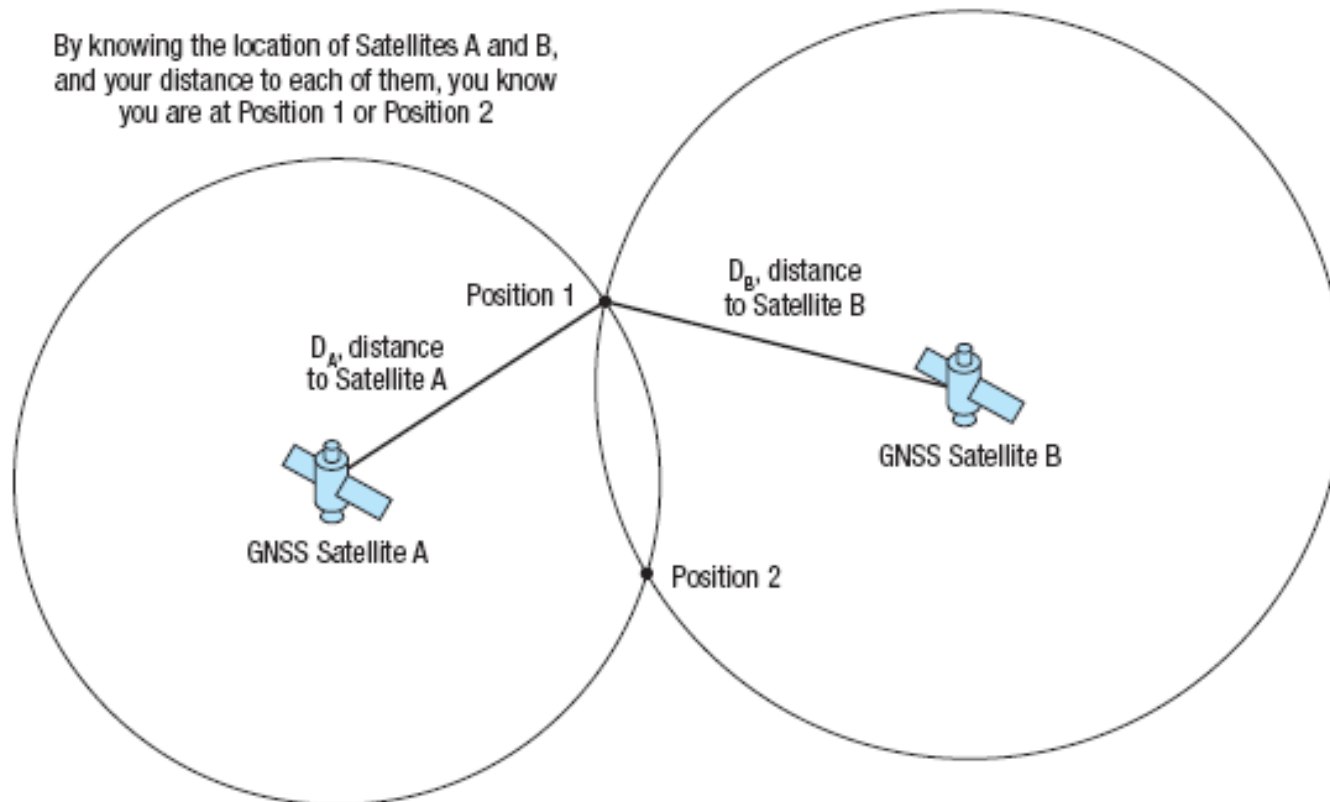


- Transmission of a pseudorandom code from a satellite. The receiver can determine the time of propagation by comparing the transmit time to the receive time.

Computation

In a two-dimensional world, here is how position calculation works:

- If receiver acquires two satellites, it has two possible positions:



GNSS Satellite Systems

- Currently, the following GNSS systems are operational
 - GPS (United States)
 - GLONASS (Russia)
 - Galileo (European Union)
 - BeiDou (China)
- The following regional navigation satellite systems are planned and are in varying stages of development:
 - IRNSS (India)
 - QZSS (Japan)

GPS

- GPS (Global Positioning System) or NAVSTAR, as it is officially called, the first GNSS system
- Launched in the late 1970's and early 1980's for the US Department of Defense
- Several generations, referred to as “Blocks”, of GPS satellites have been launched
- GPS was initially launched for military use, but opened up to civilian use in 1983



GPS

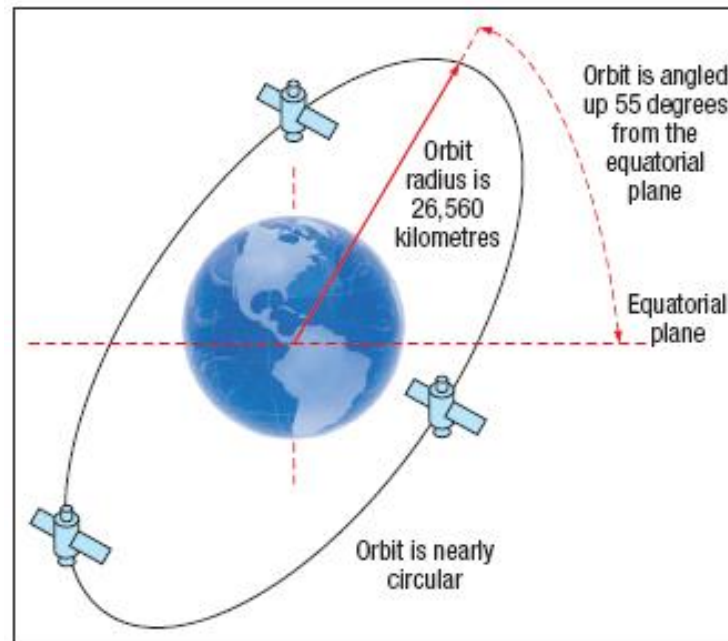
- The GPS space segment is summarized in the table below:

Satellites	21 plus 3 spares
Orbital planes	6
Orbit inclination	55 degrees
Orbit radius	26,560 km

- The orbital period of each satellite is approximately 12 hours
- At any point in time, a GPS receiver will have at least 6 satellites in view at any point on Earth under open sky conditions

GPS

- GPS orbits approximately 26,560 km above the Earth center
- GPS satellites continuously broadcast their identification, ranging signals, satellite status and corrected ephemerides (orbit parameters)
- Each satellite is identified by their Space Vehicle Number (SVN) and their PseudoRandom code Number (PRN)



GPS

- GPS Control Segment consists of a master control station and a backup master control station, in addition to monitor stations throughout the world



- Monitor stations track the satellite broadcast signal and pass them on to the master control stations where the ephemerides are recalculated. The resulting ephemerides and timing corrections are transmitted back to the satellites through data up-loading stations

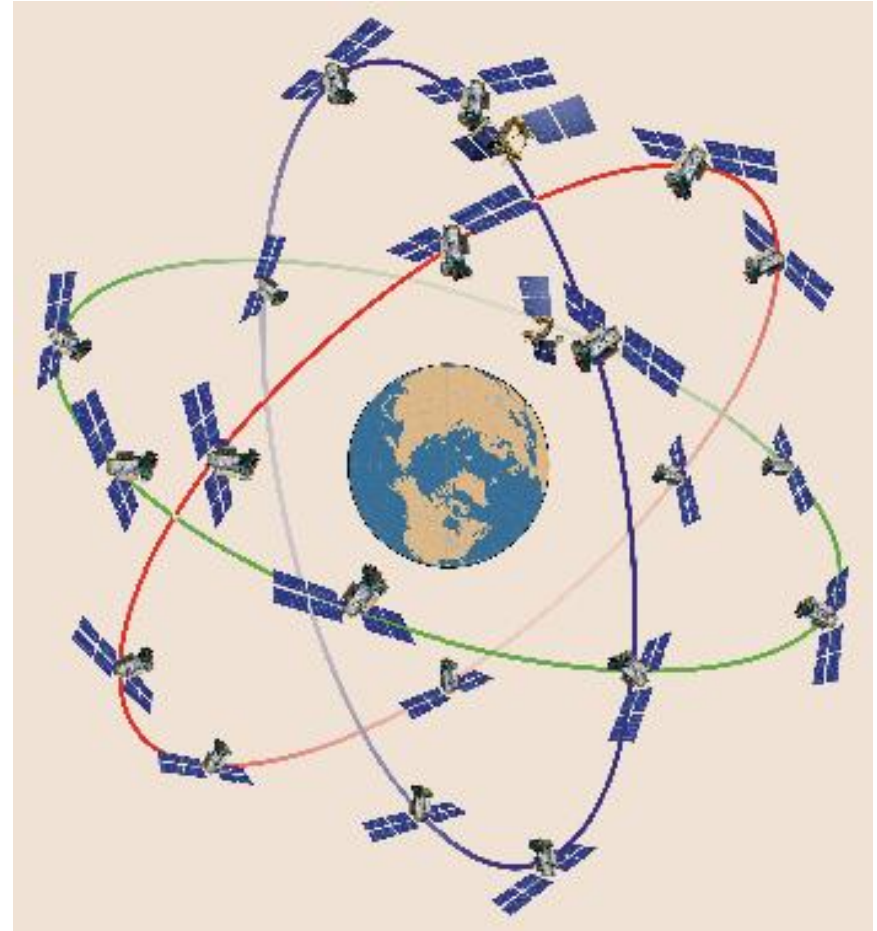
GLONASS

- GLONASS (Global Navigation Satellite System) was developed by the Soviet Union as an experimental military communications system during the 1970s
- When the Cold War ended, the Soviet Union recognized that GLONASS can be used in commercial applications
- First satellite was launched in 1983, and system declared fully operational in 1993
- GLONASS went through a period of performance decline
- Russia is committed to bring the system back up to operational and set a date of 2011 for full deployment of the system



GLONASS

- The GLONASS constellation provides visibility to a variable number of satellites, depending on your location
- The GLONASS space segment consists of 24 satellites in three orbital planes
- The GLONASS constellation geometry repeats about once every eight days
- GLONASS satellites orbit 25,510 km above the Earth's surface. About 1,050 km lower than GPS satellites



Galileo

- Europe's global navigation system
- Guaranteed global positioning service under civilian control
- Guaranteed availability of service under all but the most extreme circumstances
- Suitable for applications where safety is crucial, such as air and ground transportation
- GIOVE-A and GIOVE-B test satellites are already in orbit



Galileo

- 3 Orbital planes
- 27 Satellites + 3 Spares
- 56° Inclination Angle
- Altitude 23,616km

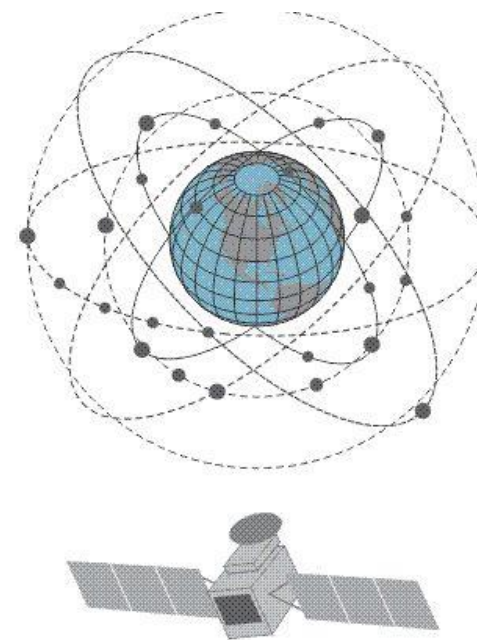
Satellites	27 operational and three active spares
Orbital planes	3
Orbital inclination	56 degrees
Orbit radius	23,616 km

BeiDou

- China's global navigation system
- Initial system will provide regional coverage
- A target of 2015 to begin implementation of GEO and MEO satellites for global coverage

Satellites	35, a combination of 5 GEO and 30 MEO
Orbital planes	6
Orbital inclination	55 degrees
Orbit radius	27,528 km

- Compass will provide two levels of services:
 - Public service for civilian use, and free to users in China
 - Licensed military service, more accurate than public service



BeiDou

6 Orbital planes

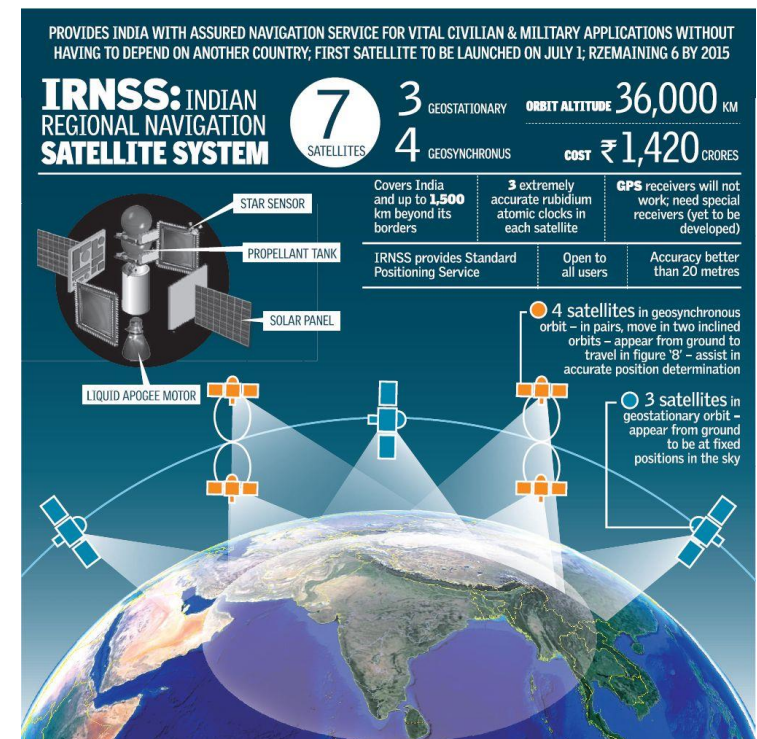
35 Satellites + 5 GEO + 27 MEO + 3 IGSO

55° Inclination Angle

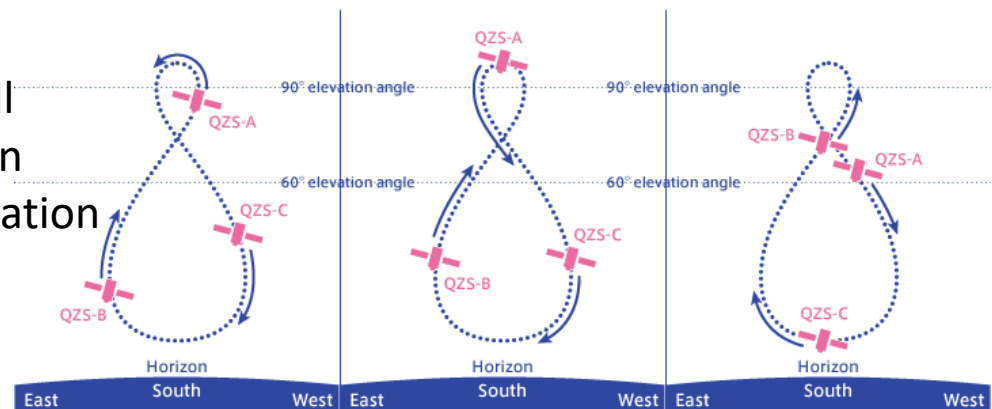
Altitude 38,300 km, 21,500 km

New GNSS Systems

- IRNSS (India Regional Navigation Satellite System, India)
 - Satellite system to provide regional coverage
 - Launched in 2013
- QZSS (Quasi-Zenith Satellite System, Japan)
 - A three satellite system that will provide regional communication services and positioning information for the mobile environment



Quasi-Zenith Satellite (QZS) movements as seen from near Tokyo



Applications

Some common GNSS Applications include:

- Transportation
- Timing
- Machine Control
- Marine
- Surveying
- Defence
- Port Automation

Transportation

- Portable navigation devices
- Air, marine, and ground based vehicle navigation



Machine Control



NovAtel Inc.



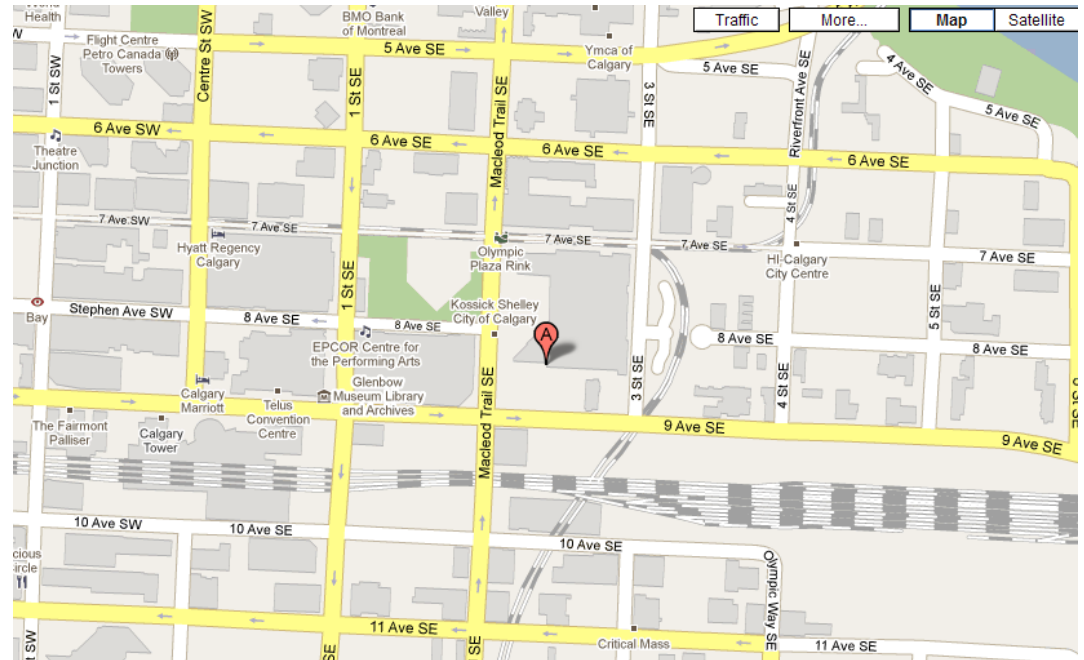
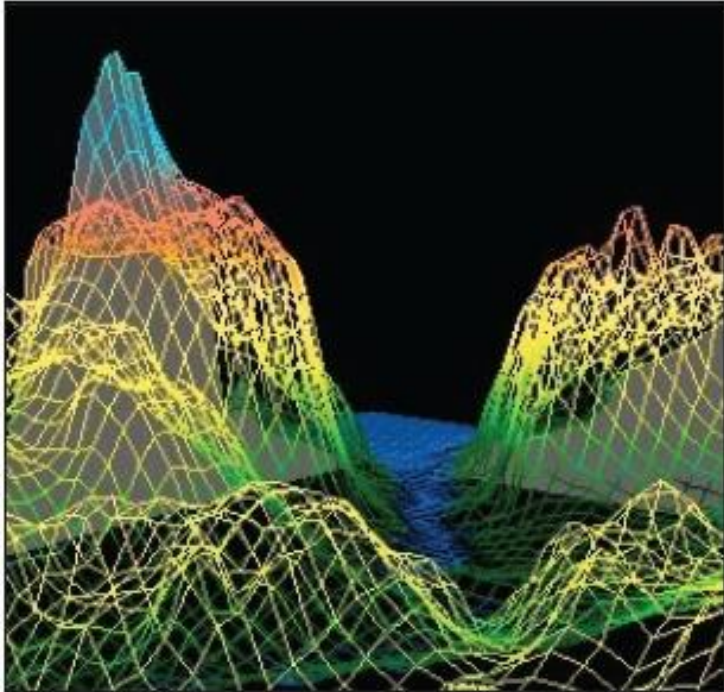
Thomas Morley, NovAtel Inc.

Surveying



Google Street View

GIS



Google Map

Port Automation

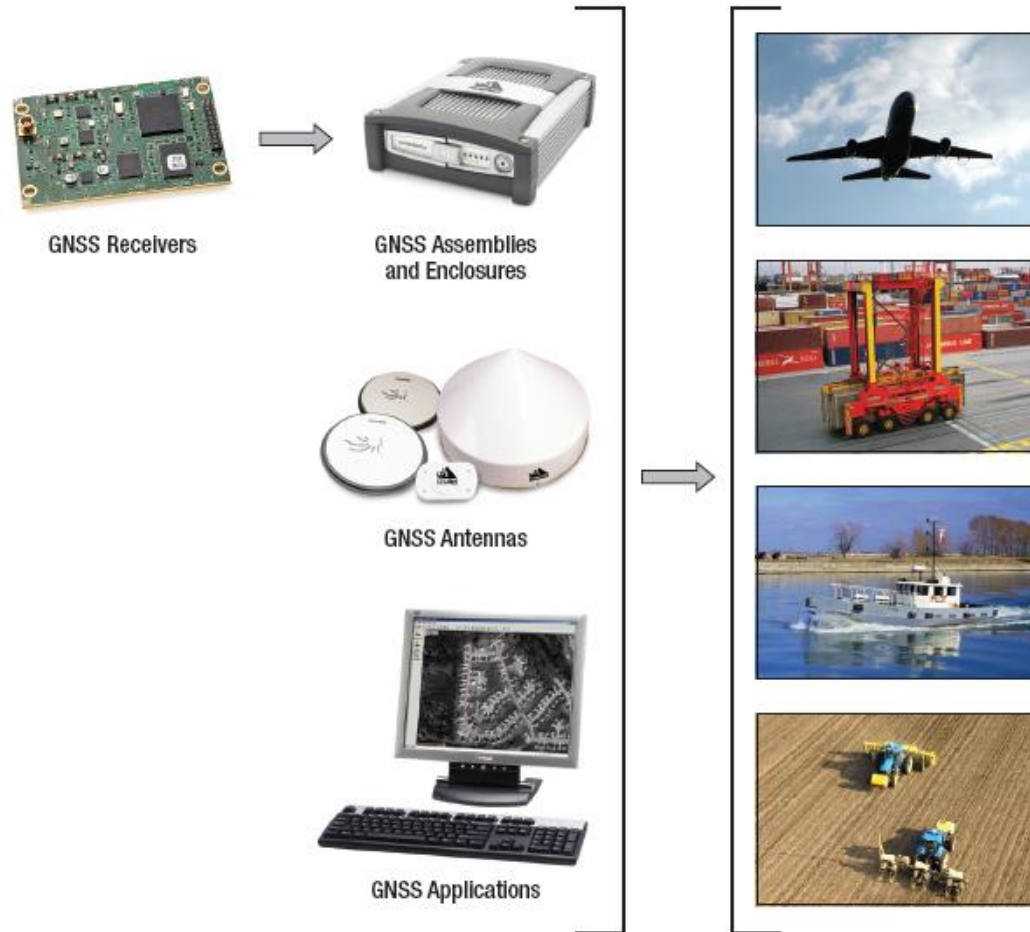


Defence



Equipment

- There are applications



ng on the

4. Remote sensing

- Next course

6. Practices and Midterm presentation

1. Field measurement

- Mapping VAST campus: collecting coordinates and information to create different layers. Ex. People density.

2. Midterm

- Team working: 4 students/ team
- Using GPS test apps.
- Building as much as possible layers, maps as you can.
- Making maps from your layers.

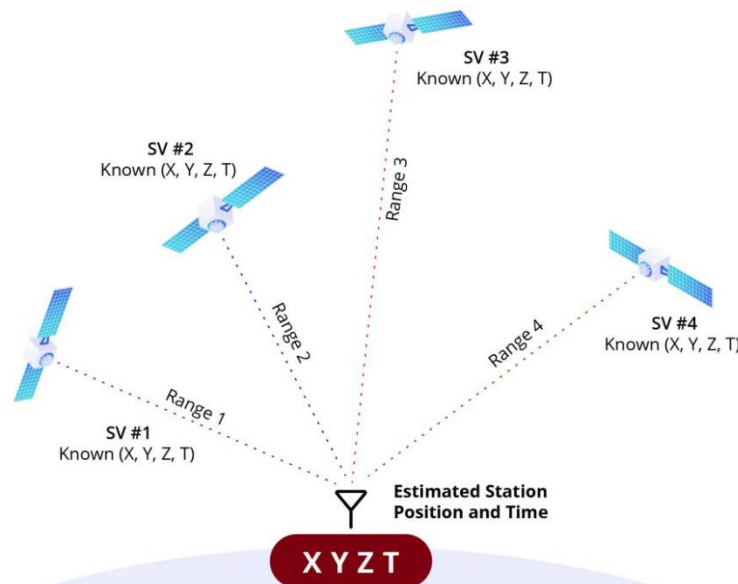
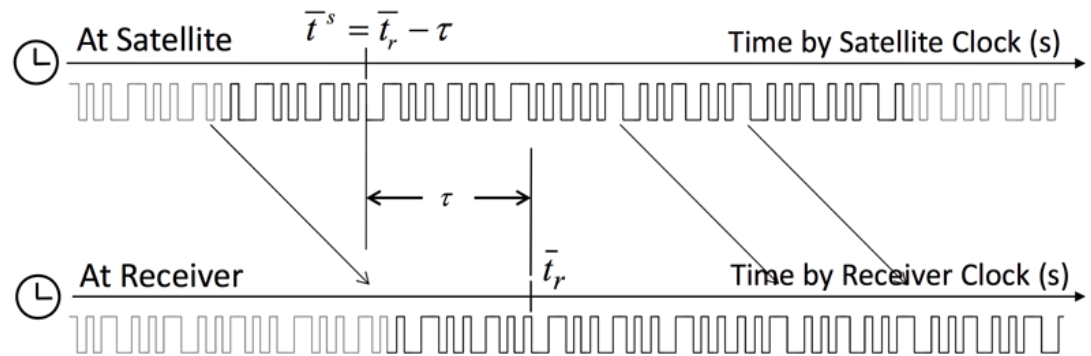
3. Presentation

- Structure: Introduction, Data and Methods, Results, Discussions, Conclusions

GNSS technique

- GNSS measurement**

Carrier phase L band (1 – 2 GHz,
Wave length 30 – 15 cm) to
measure distance



The Global Positioning System

Measurements of code-phase arrival times from at least four satellites are used to estimate four quantities: position in three dimensions (X, Y, Z) and GNSS time (T).

Instruction for practices

- **Install: GPS Test app (Android)**

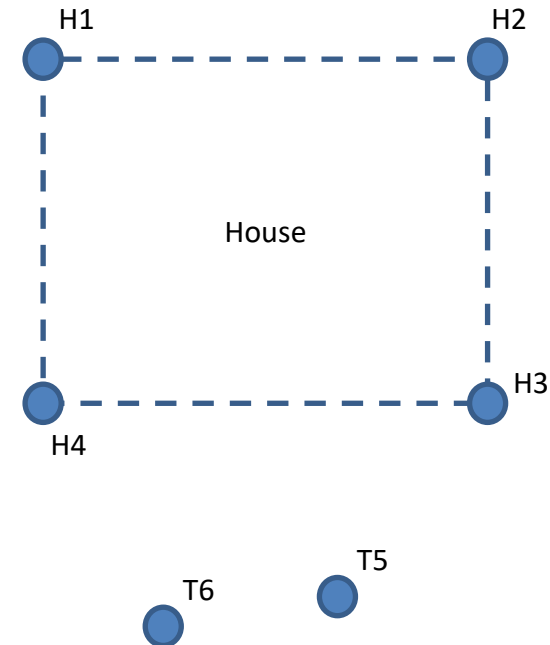
GPS Diagnostic or Tomstrails GPS UTM
(IOS)



Transfer to the QGIS

**Coordinates measured by GNSS and type in Excel
then export to .csv format**

Name	X	Y	H	Description
H1	5546225	2456823	15	House (A3, 20 m)
H2	5546225	2456823	15	House (A3, 20 m)
H3	5546225	2456823	15	House (A2+ Road)
H4	5546225	2456823	15	
T5	5546225	2456823	15	Tree (Mango 10 m)
...				



In QGIS

Layer /Add layer/ Add delimited text

Remember to indicate the coordinate system, x, y, h.